

Characterising sensorimotor representations with multivariate pattern analysis

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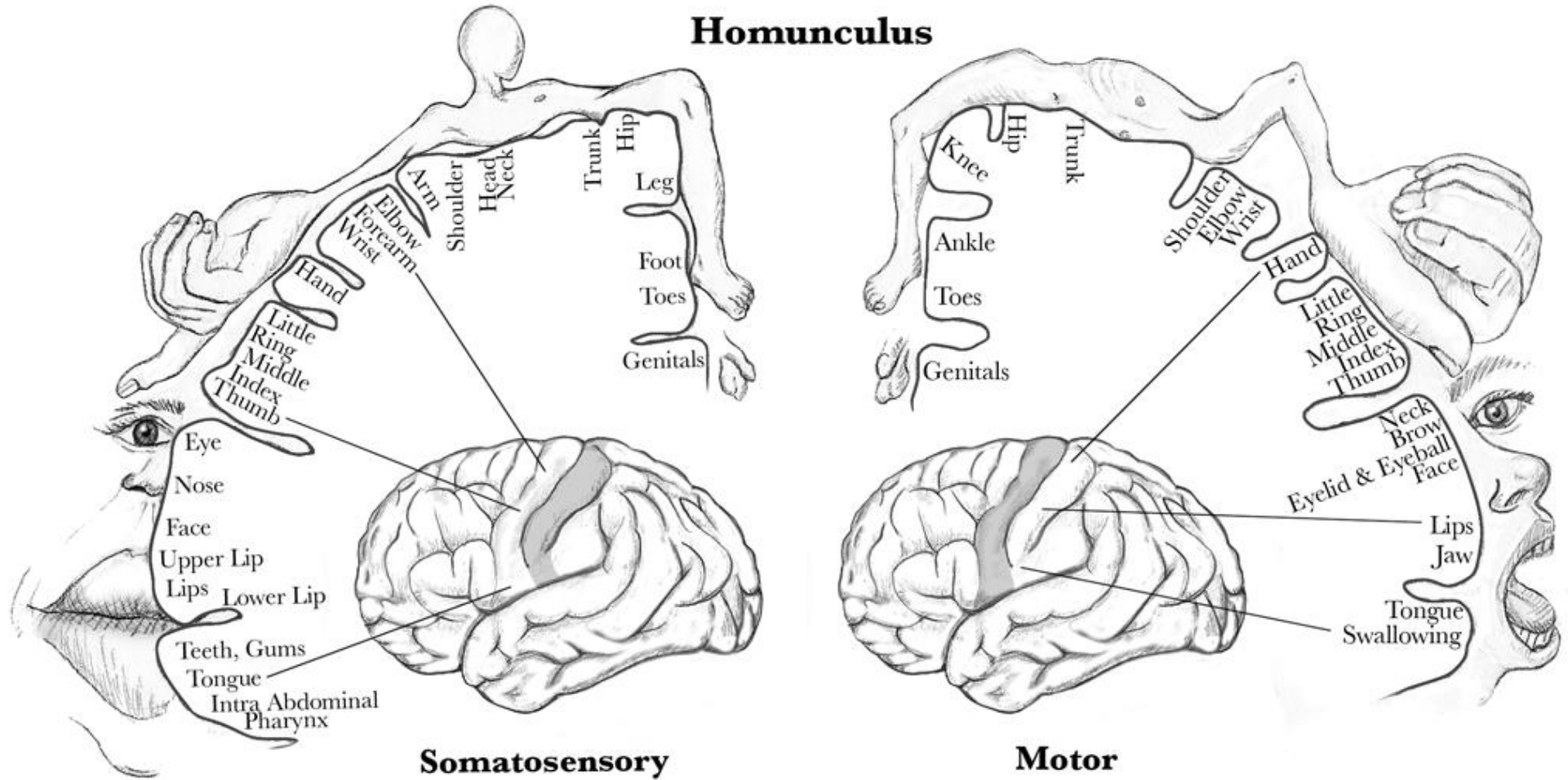
17 August 2026



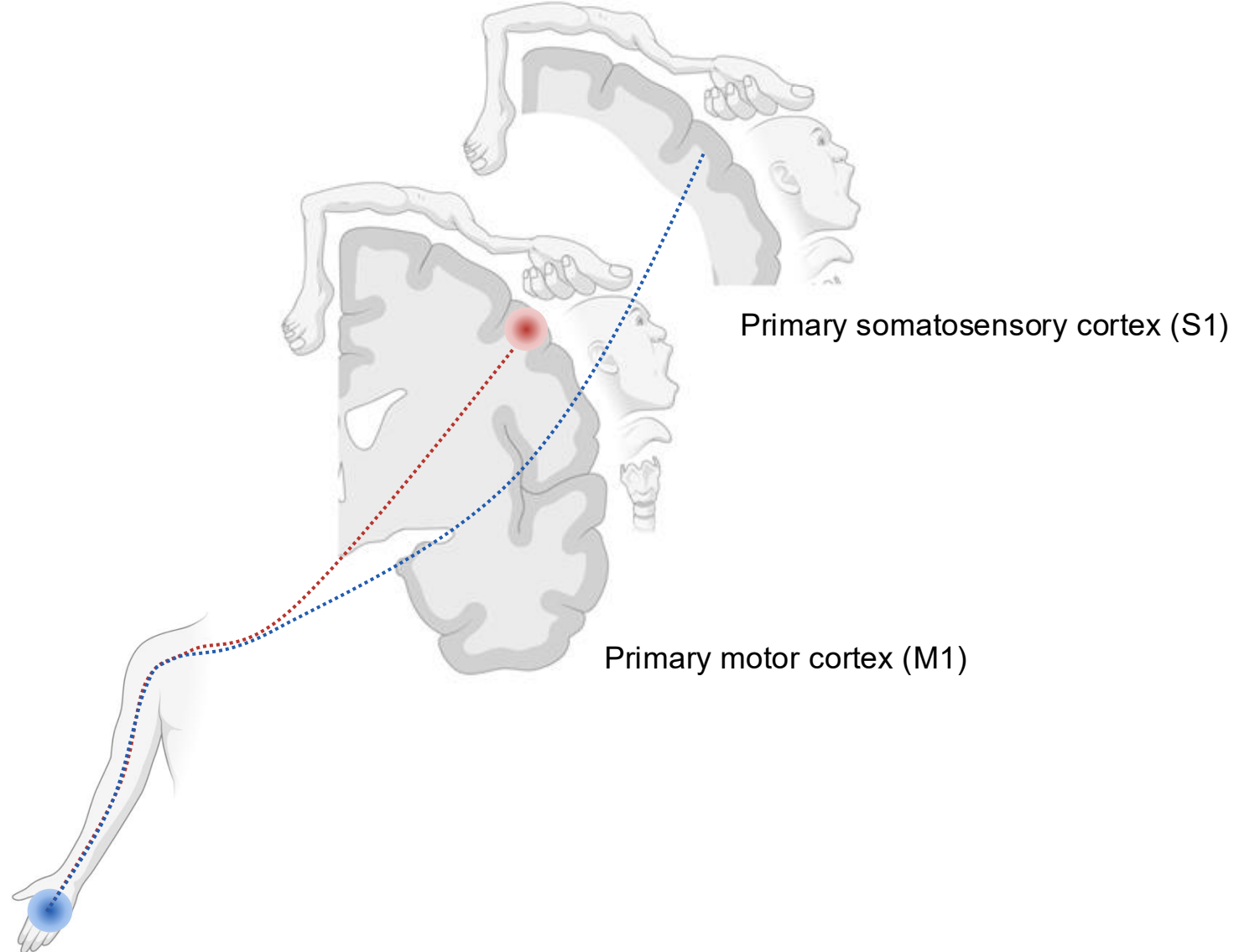
Sensorimotor functions are essential for interacting with the world



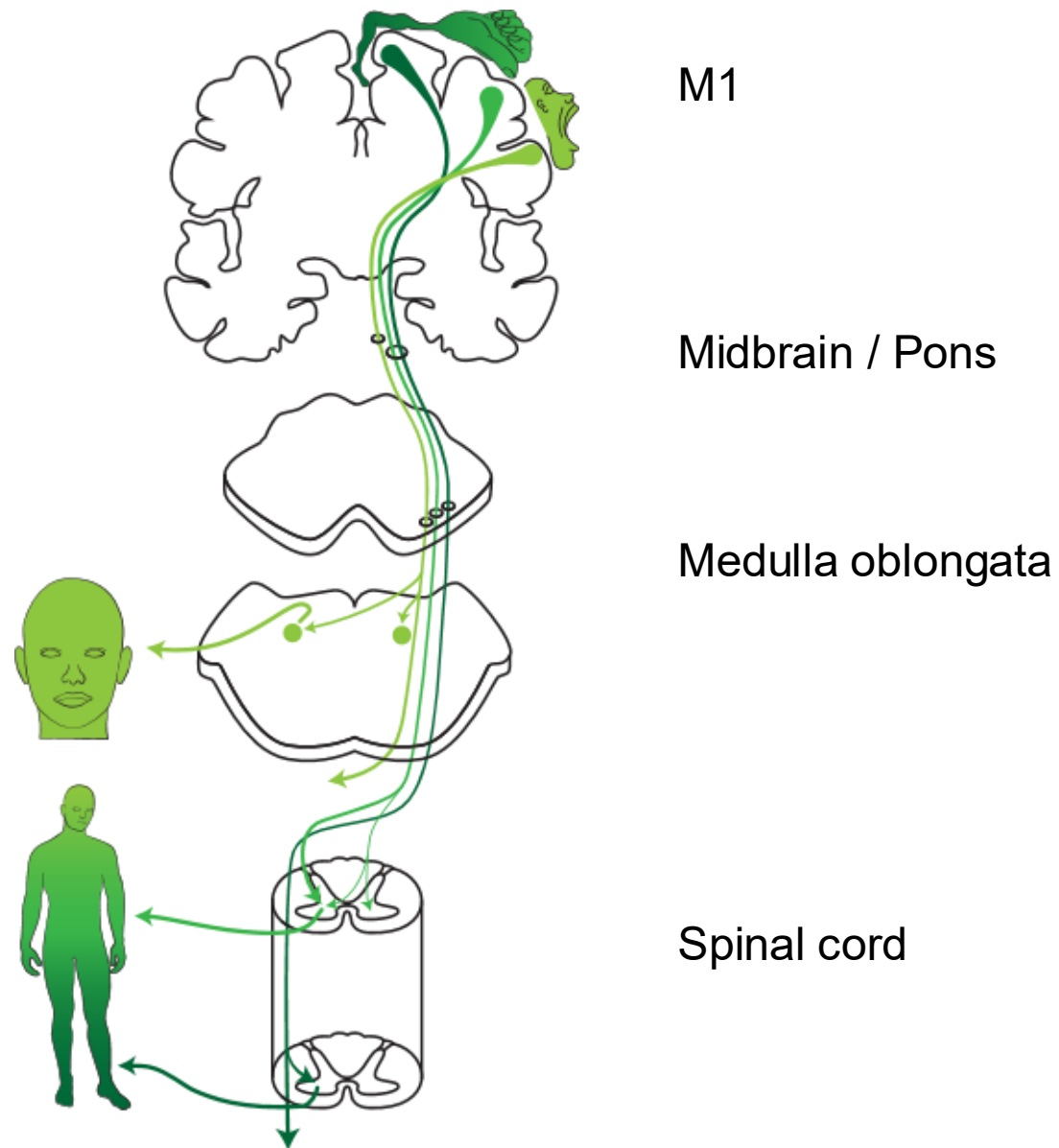
The sensorimotor system is somatotopically organised



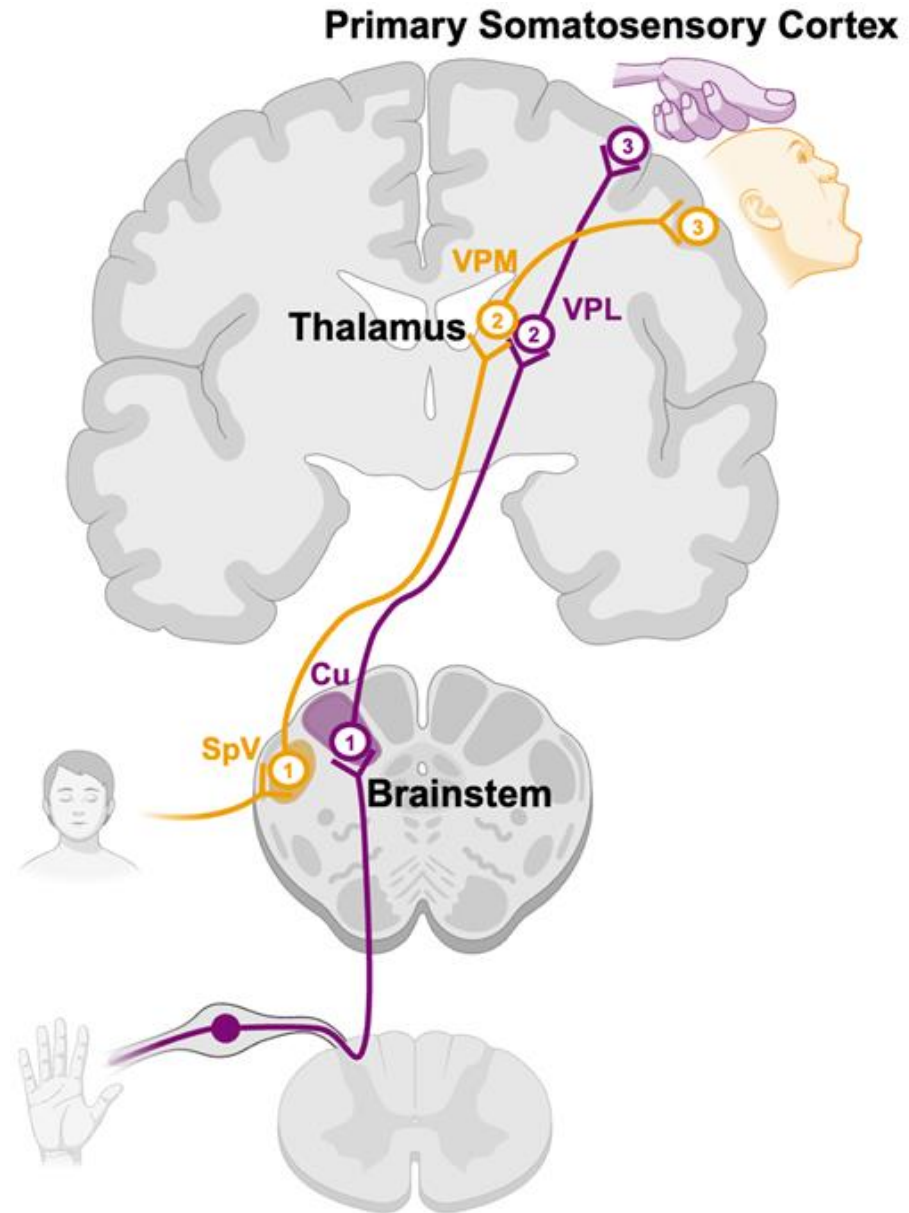
The sensorimotor system is somatotopically organised



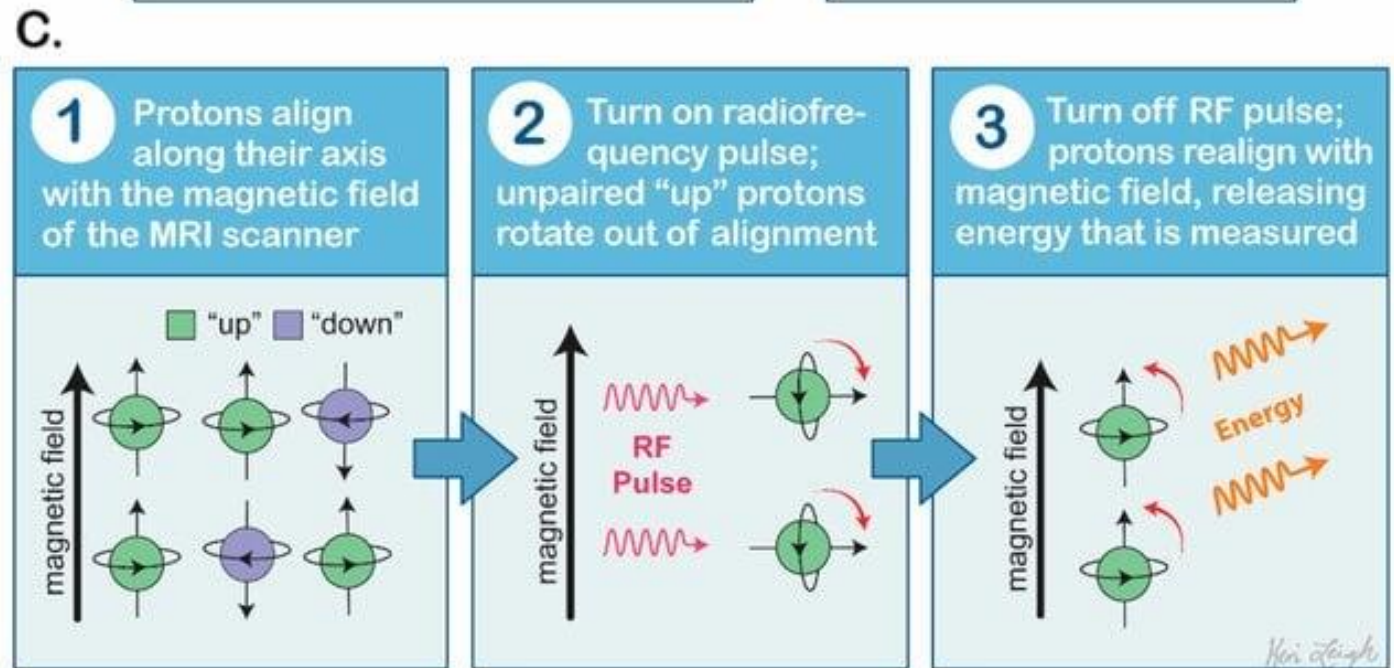
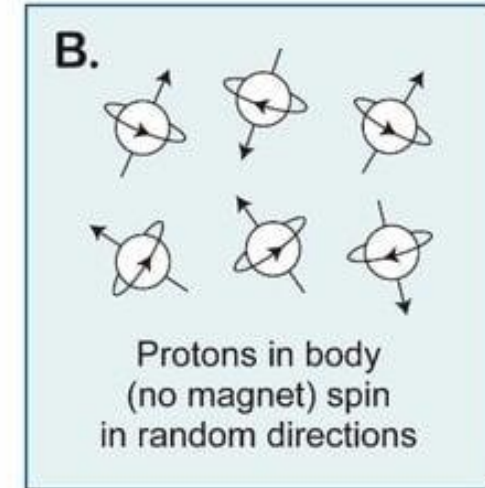
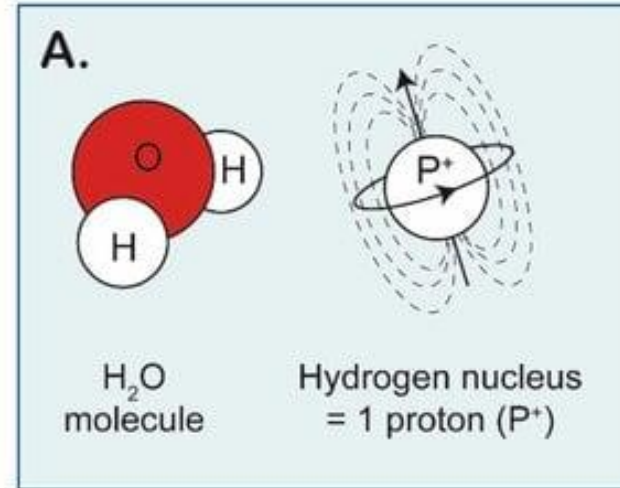
Motor control: The corticospinal tract



Somatosensory processing

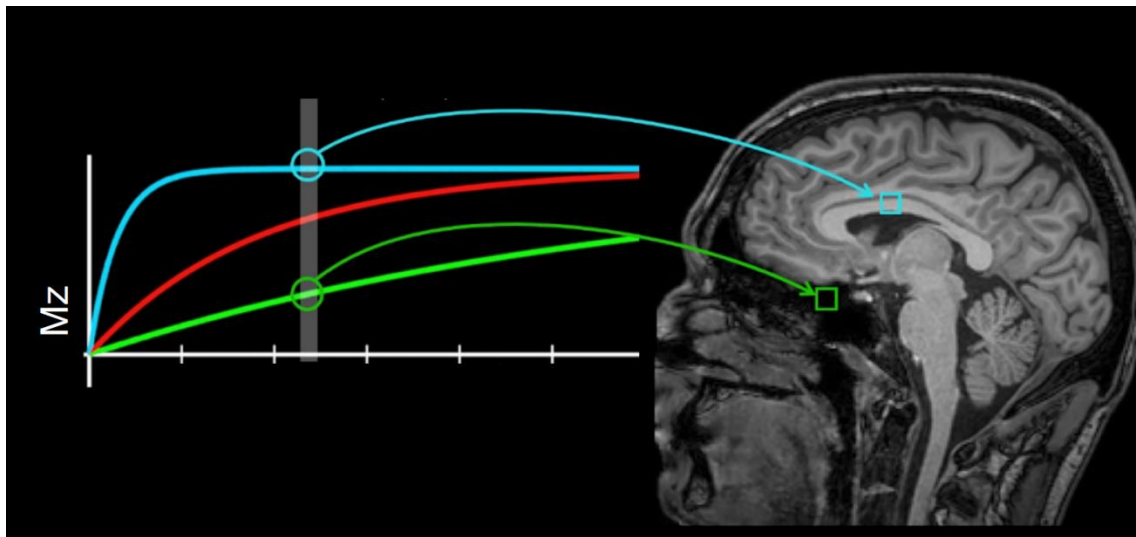


How does MRI work?

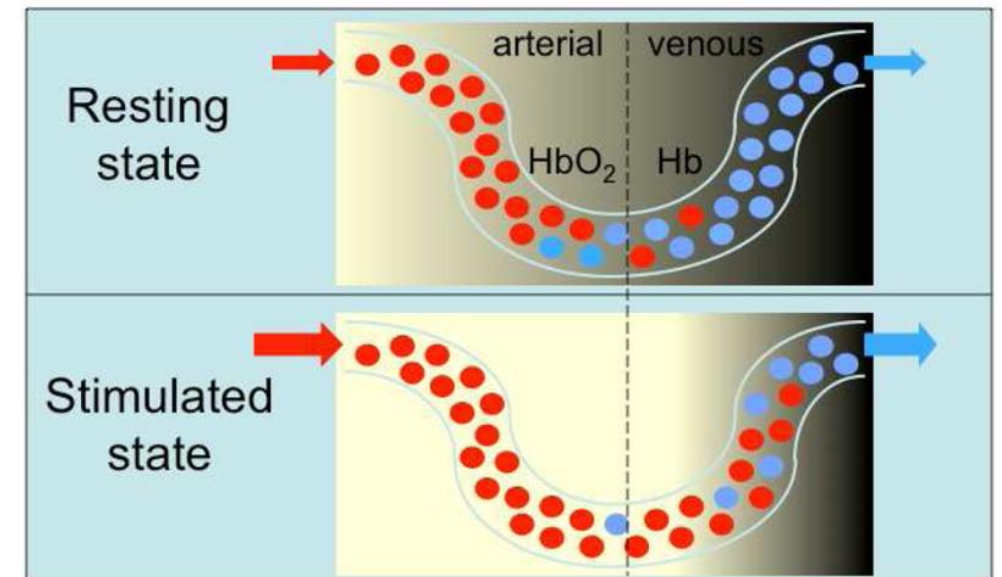


fMRI measures the Blood Oxygen Level Dependent (BOLD) signal

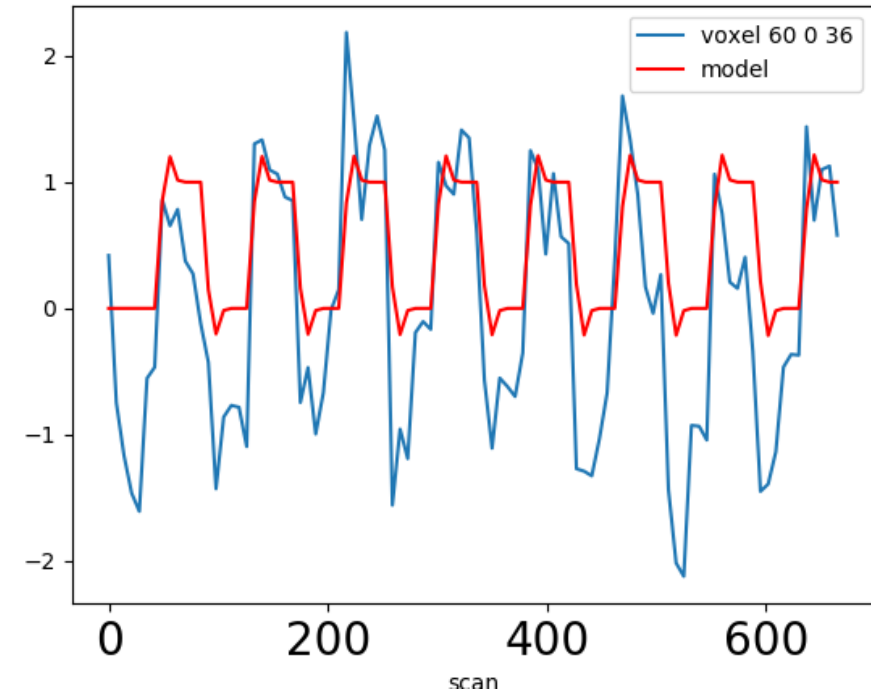
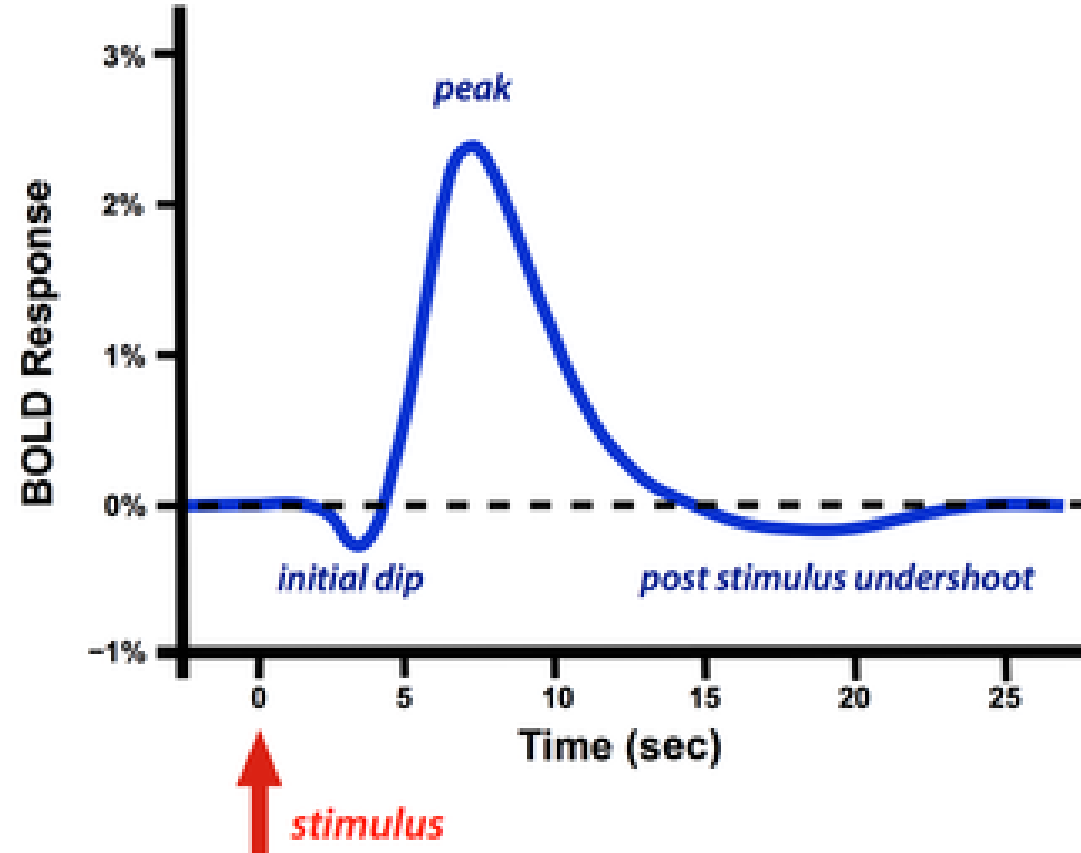
Realignment of hydrogen atoms with magnetic field depends on tissue type



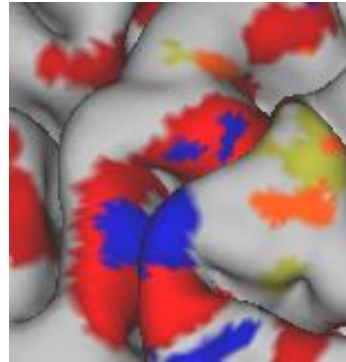
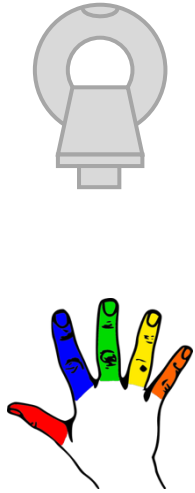
Changes in neural activity / metabolism lead to changes in cerebral blood flow (neurovascular coupling) and changes in ratio of oxygenated / deoxygenated hemoglobin



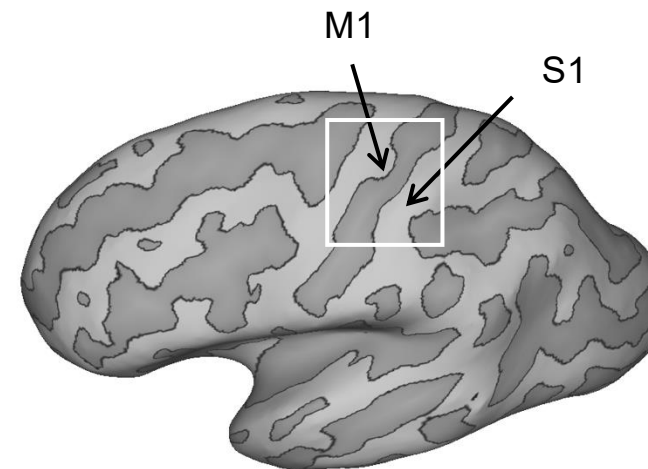
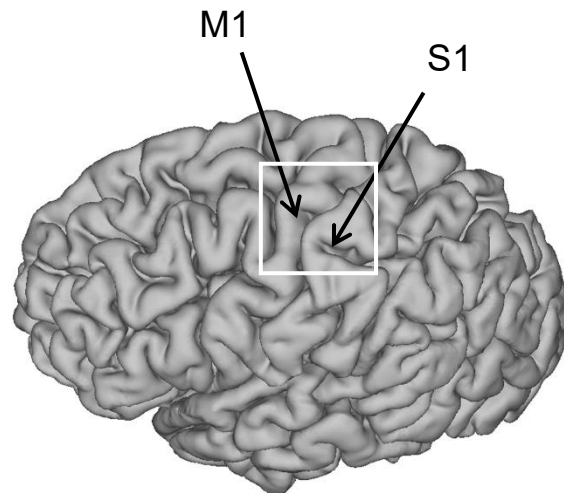
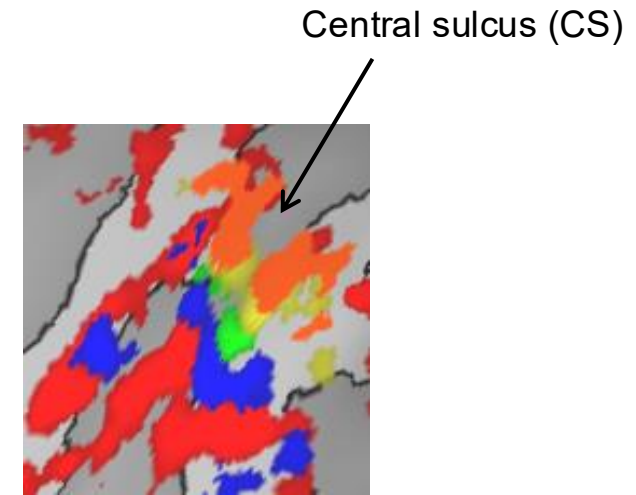
Blood Oxygen Level Dependent (BOLD) signal



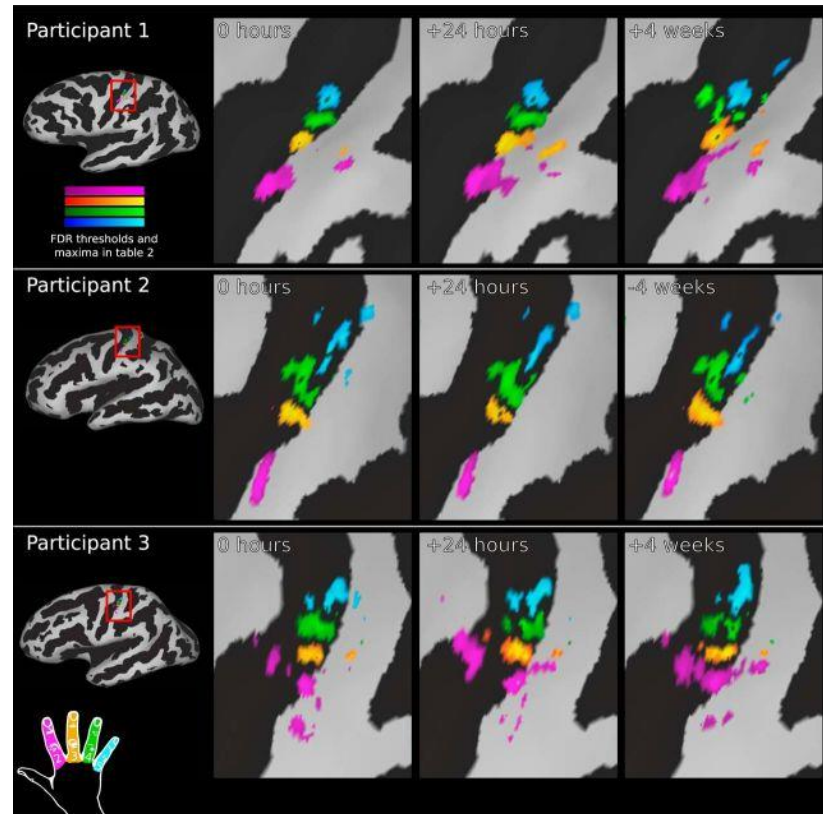
Somatotopy of finger representations



Whole



Finger maps are stable over time within participants



FDR threshold ($\alpha = 0.001$)



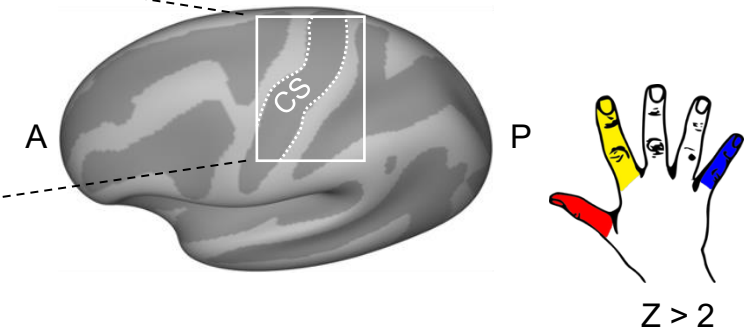
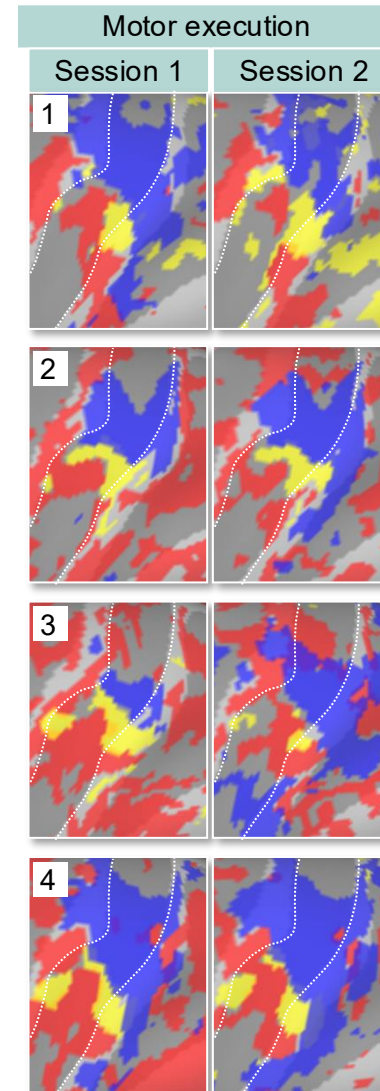
7T fMRI
2 mm³



N = 9



Kolasinski et al. (2016). JNeurosci.



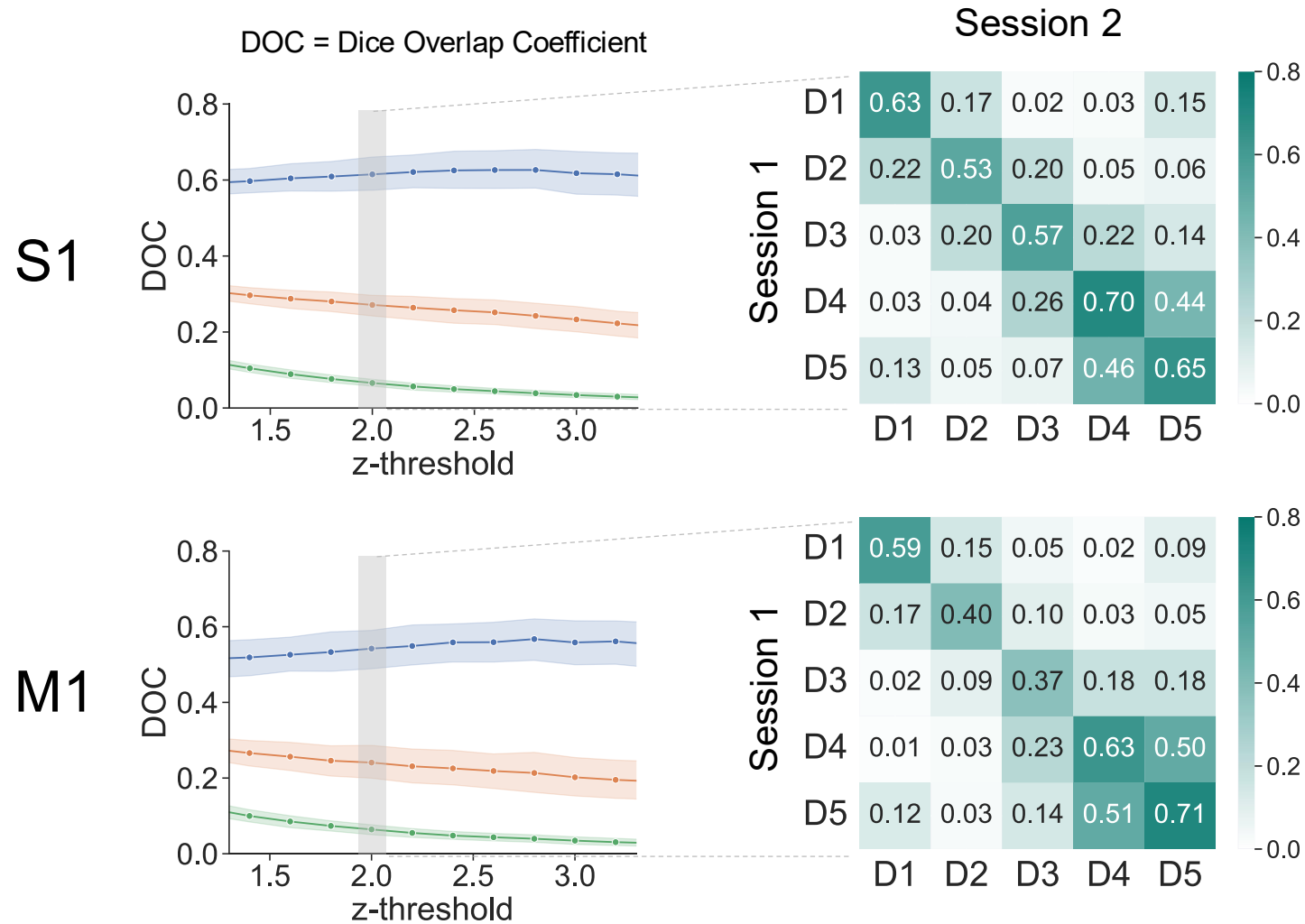
3T fMRI
2.2 mm³



N = 16

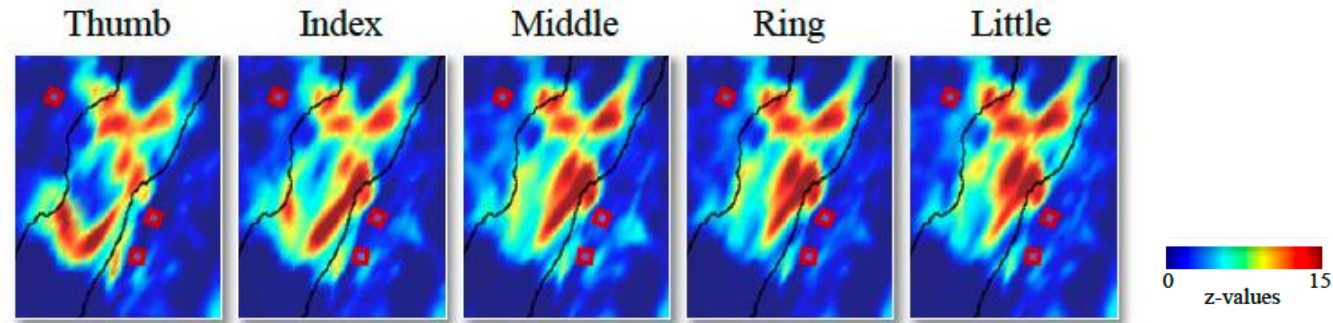
Odermatt et al. (2024). bioRxiv.

Neighbouring finger representations overlap more than non-neighbouring

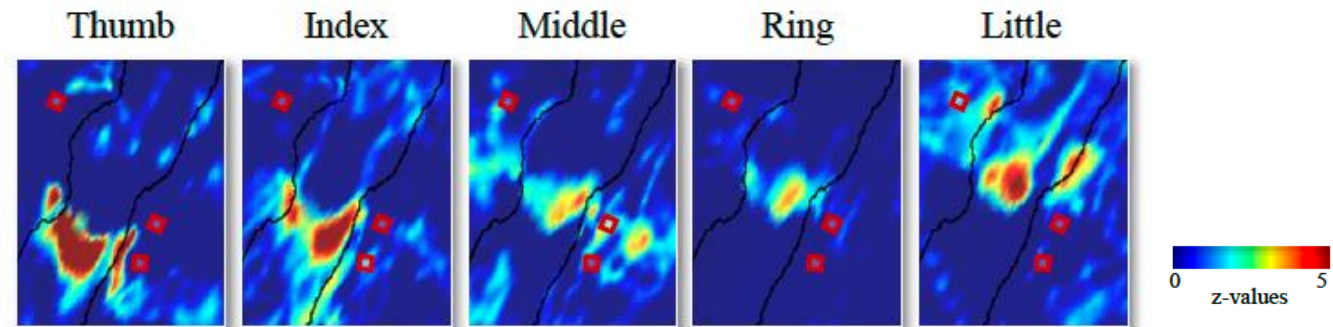


Univariate finger maps typically show **finger-selective** activity

Finger > rest

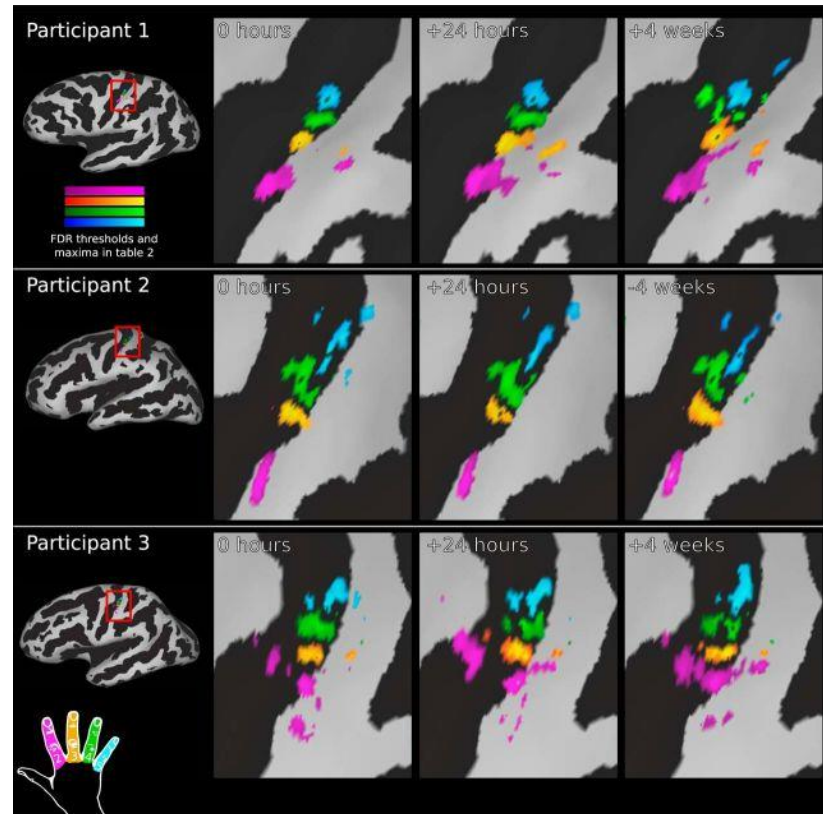


Finger-selective:
Finger > other fingers



- Finger-selective activity clusters depend on
- threshold, which is arbitrarily chosen
 - the included conditions

Finger maps are stable over time within participants



FDR threshold ($\alpha = 0.001$)



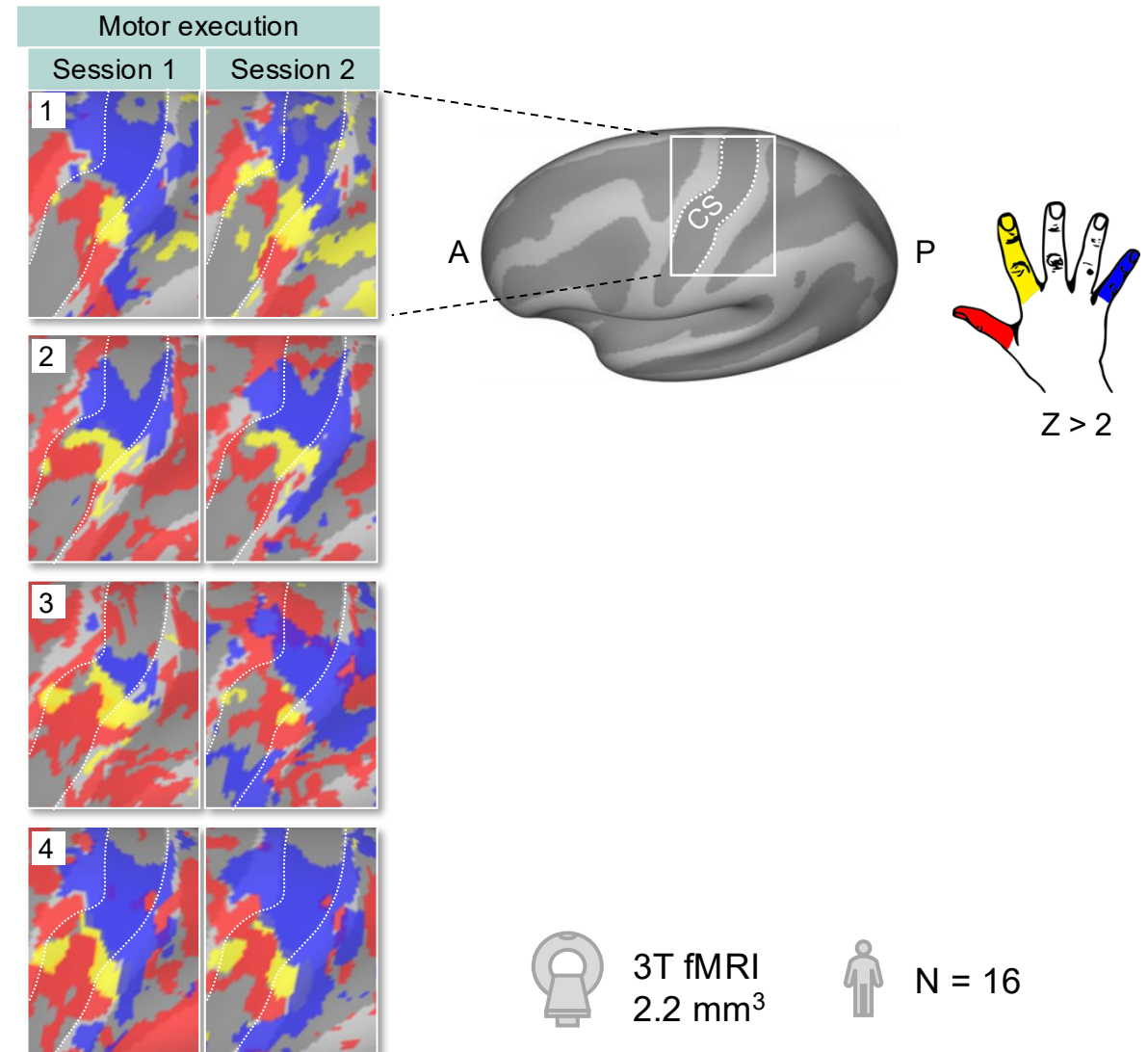
7T fMRI
2 mm³



N = 9

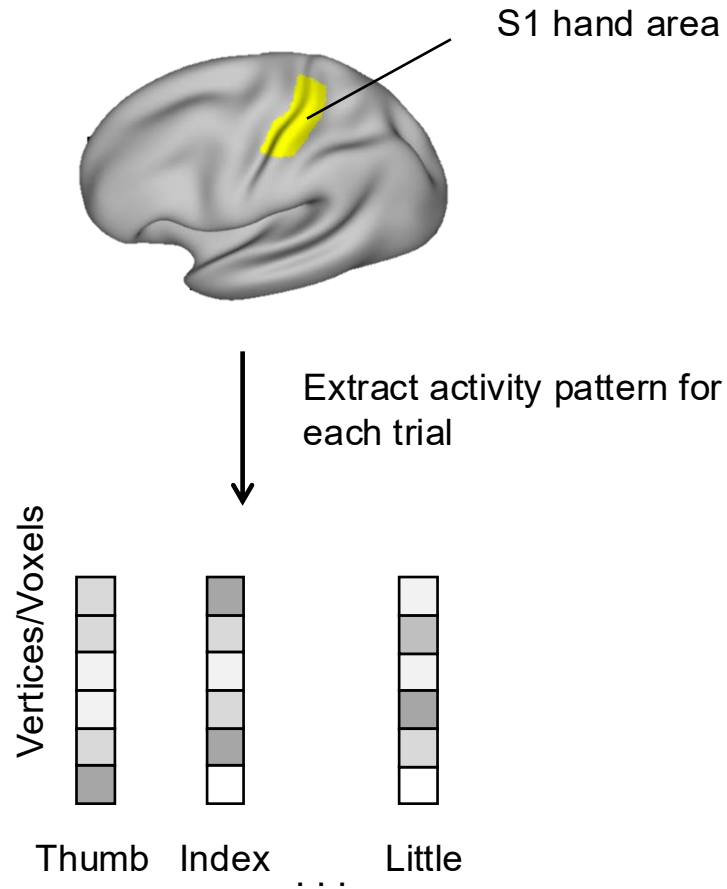


Kolasinski et al. (2016). JNeurosci.



Odermatt et al. (2024). bioRxiv.

Multivariate / Multivoxel pattern analysis (MVPA): Taking into account the whole activity pattern



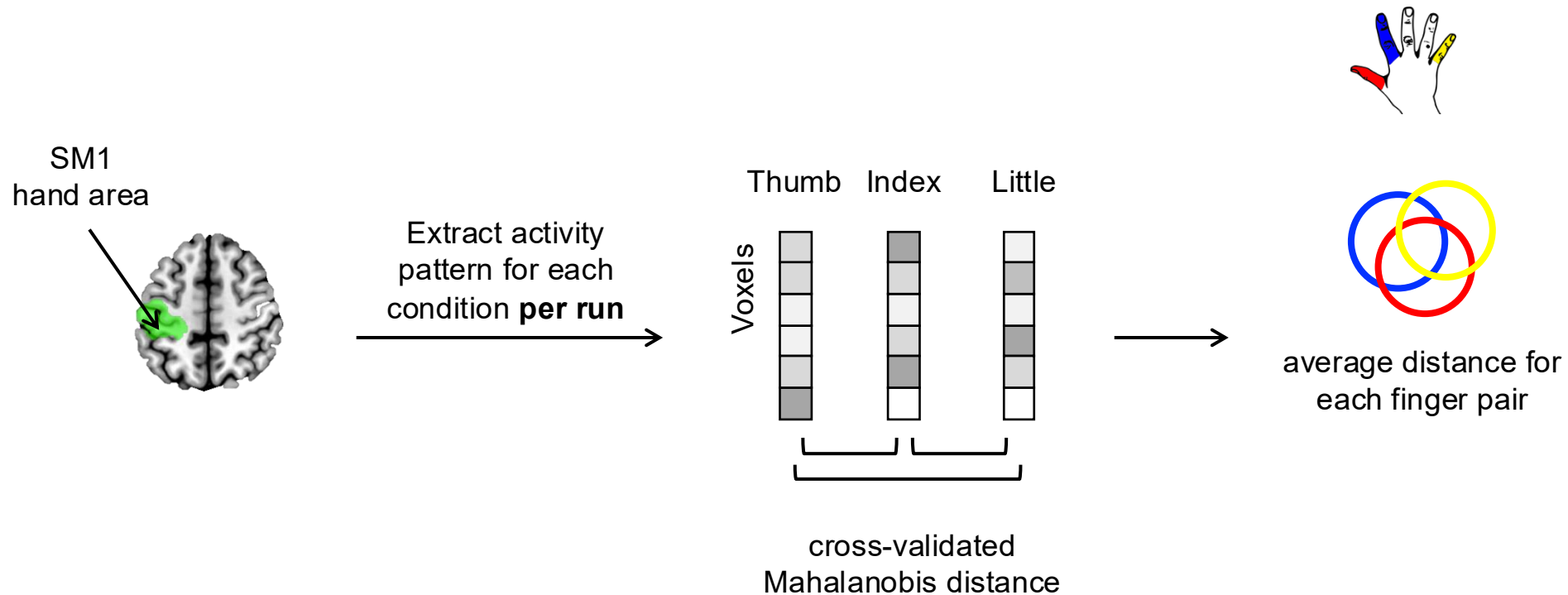
Representational similarity analysis

- Calculating distances between activity patterns

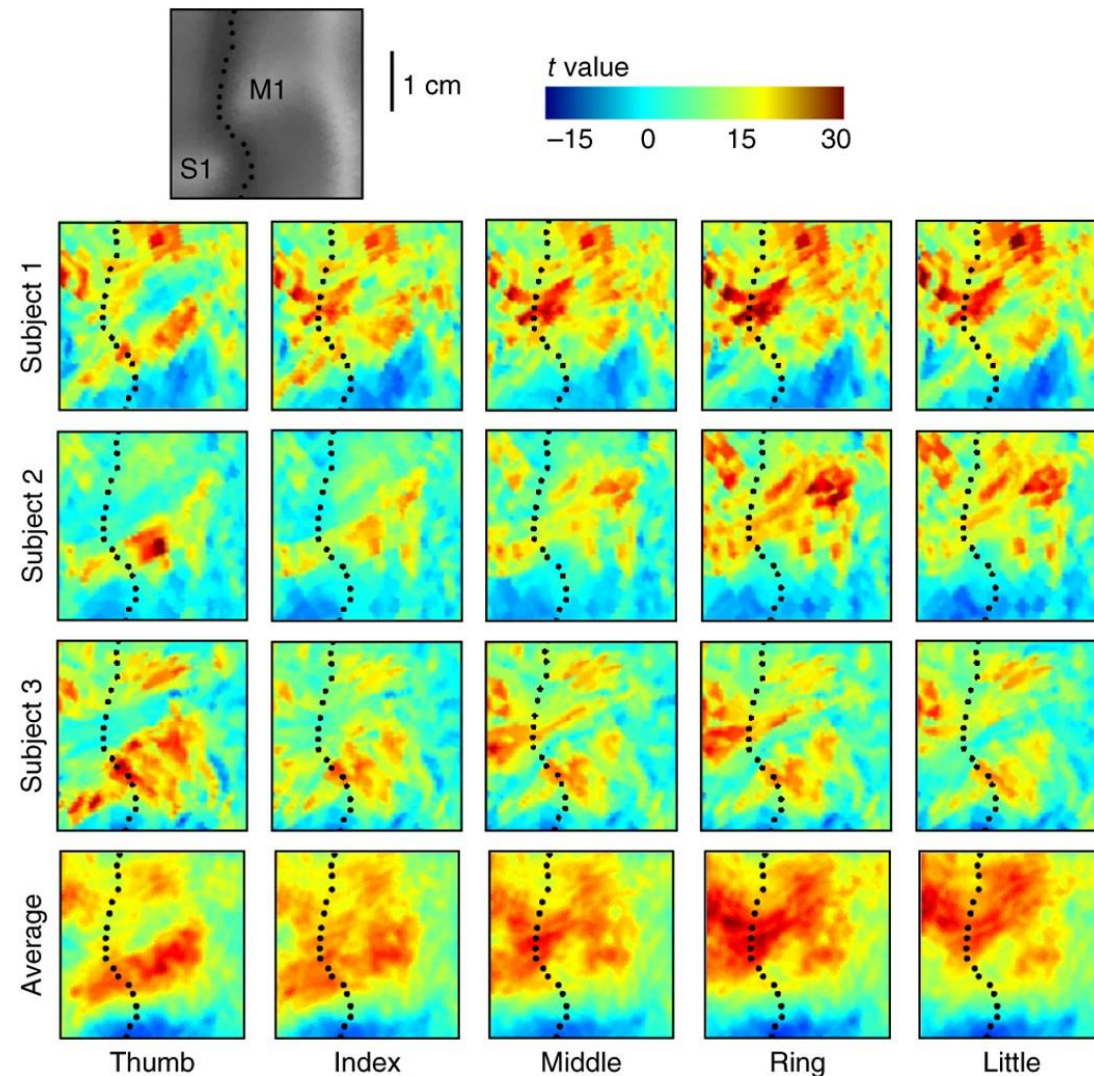
Decoding analysis

- Training a classifier to decode individual conditions based on activity patterns

Representational similarity analysis (RSA)



Exact topography of finger representations varies across participants



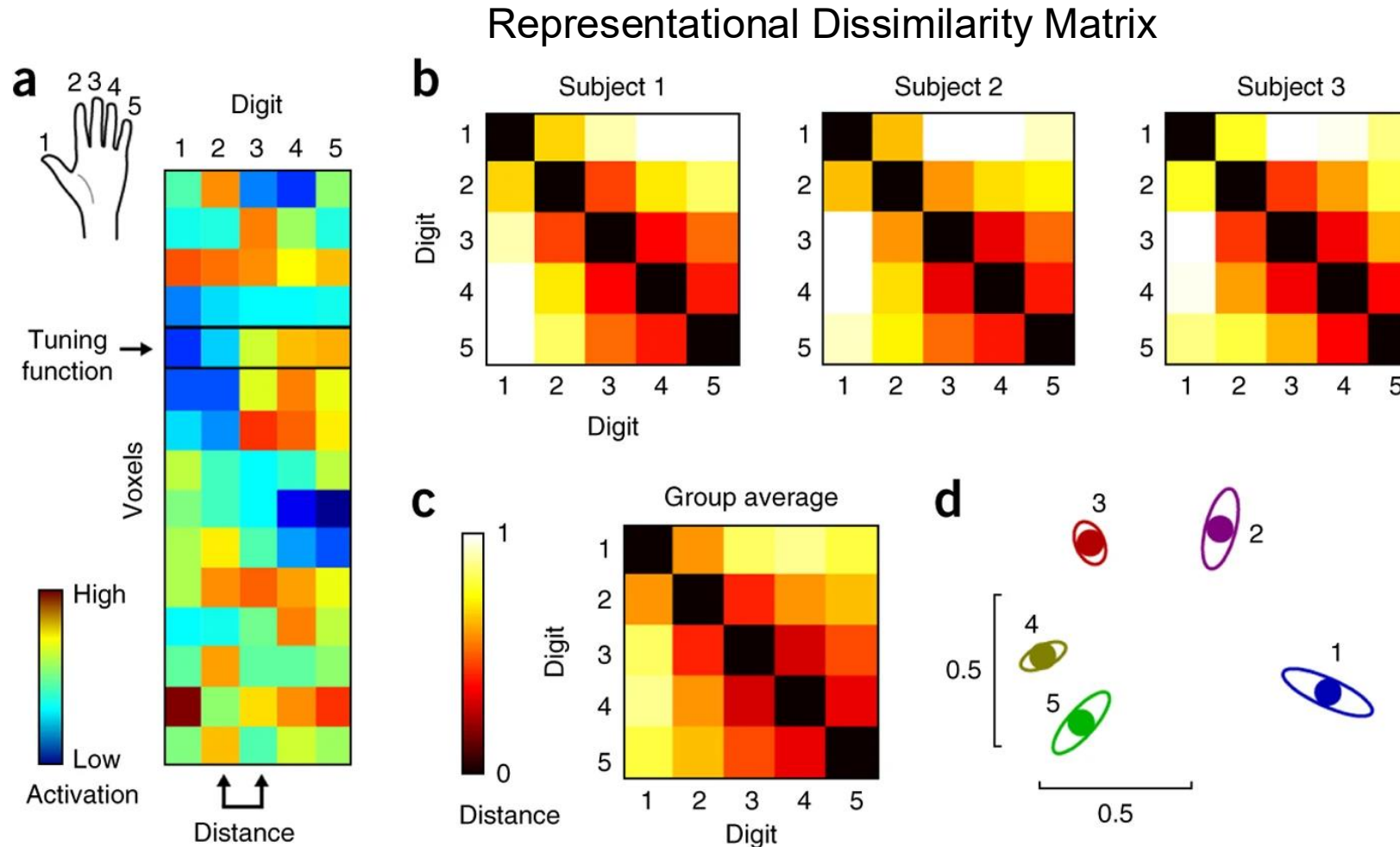
7T fMRI
1.4 mm³



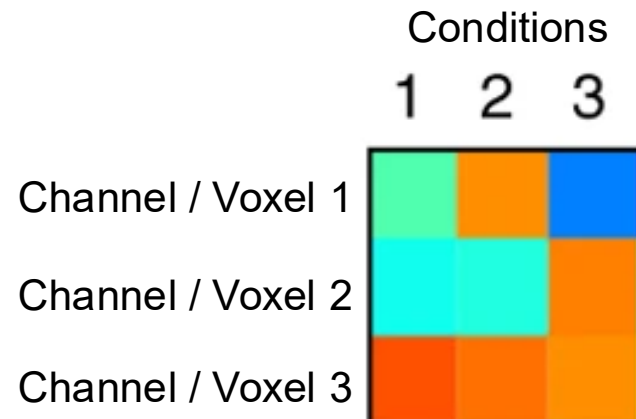
N = 7

Representational structure is invariant across participants

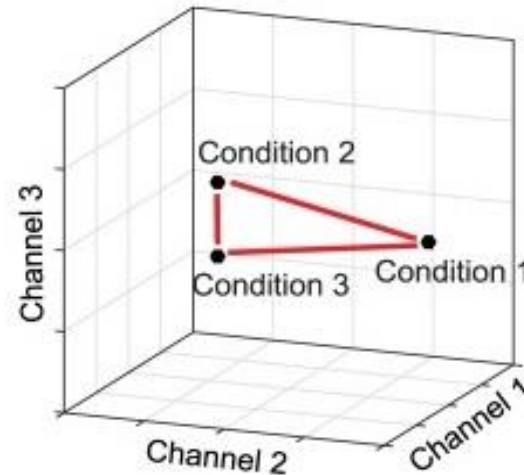
M1



Multivariate distances

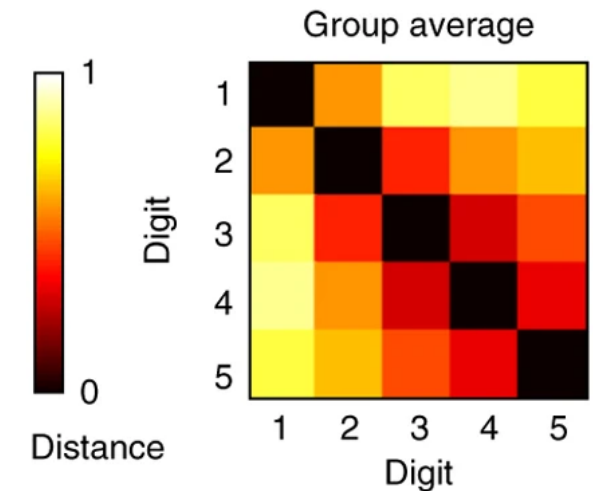


Multivariate noise normalized activity patterns (by residual covariance) per condition and run



Distance estimate:
e.g. cross-validated Mahalanobis distance

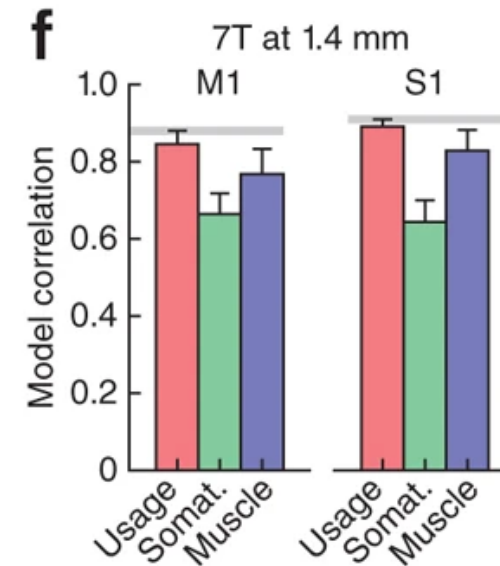
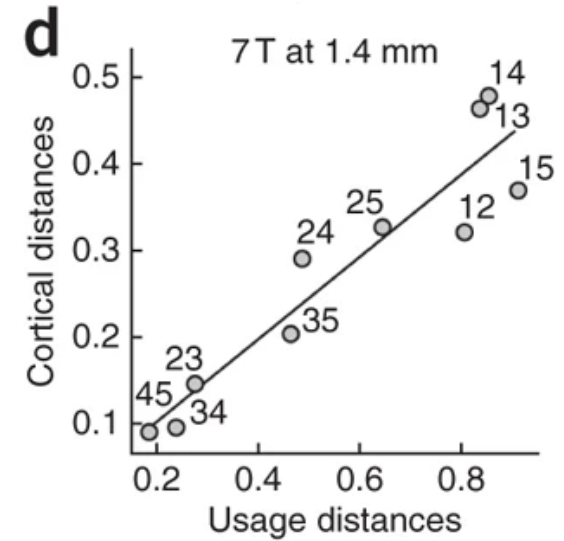
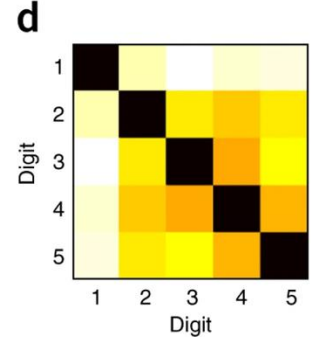
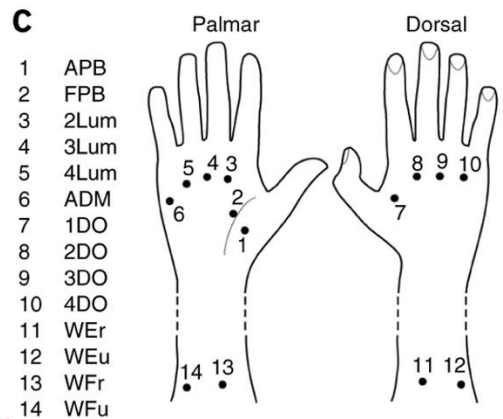
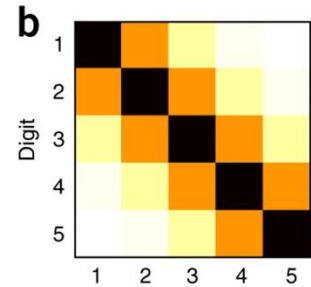
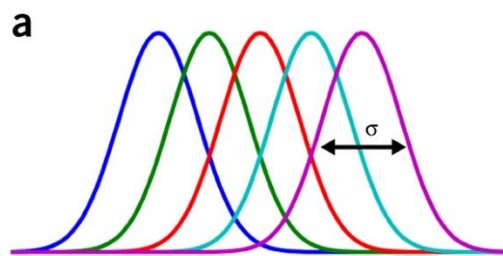
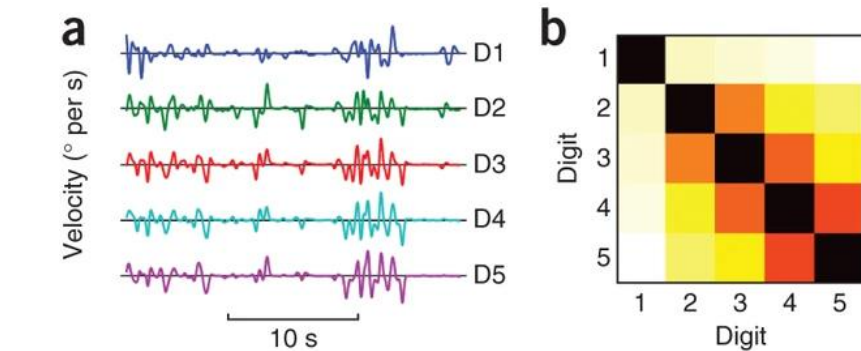
- Estimate distance vector based on n-1 runs
- Multiply with distance vector of left-out run



Representational dissimilarity matrix
i.e. representational structure

↓
if average distance at group level > 0: possible to distinguish between conditions based on activity patterns

Representational structure is predicted by hand use



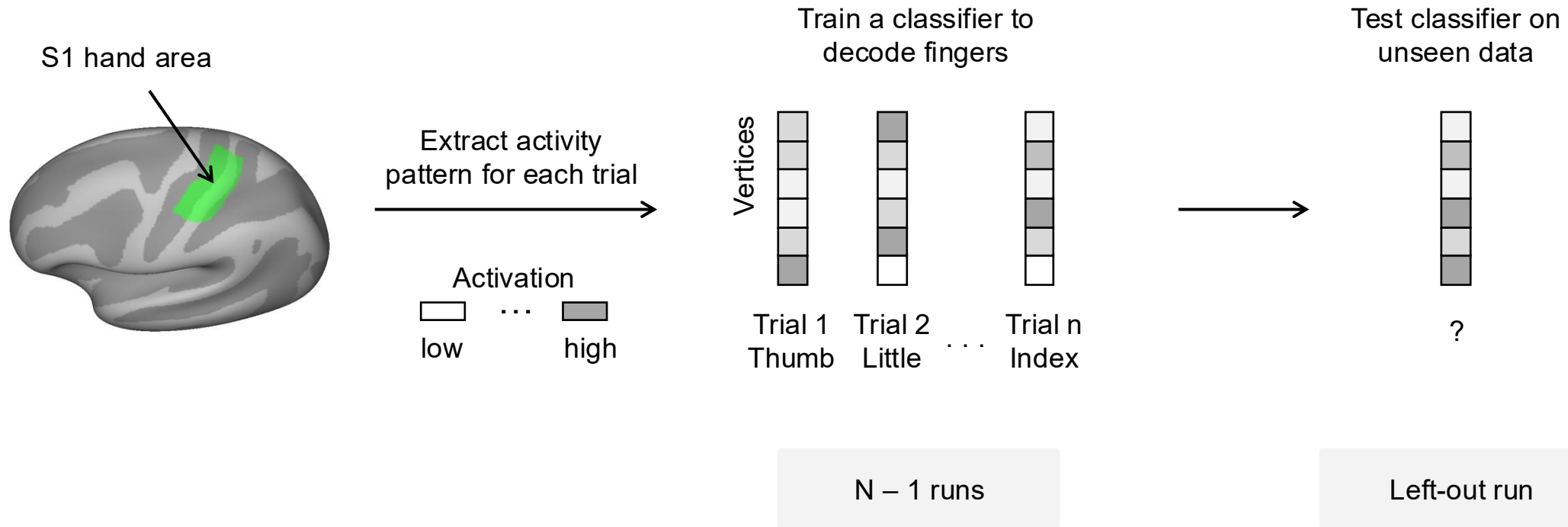
Multivariate pattern analysis

Representational similarity analysis (RSA)

How is the information organised

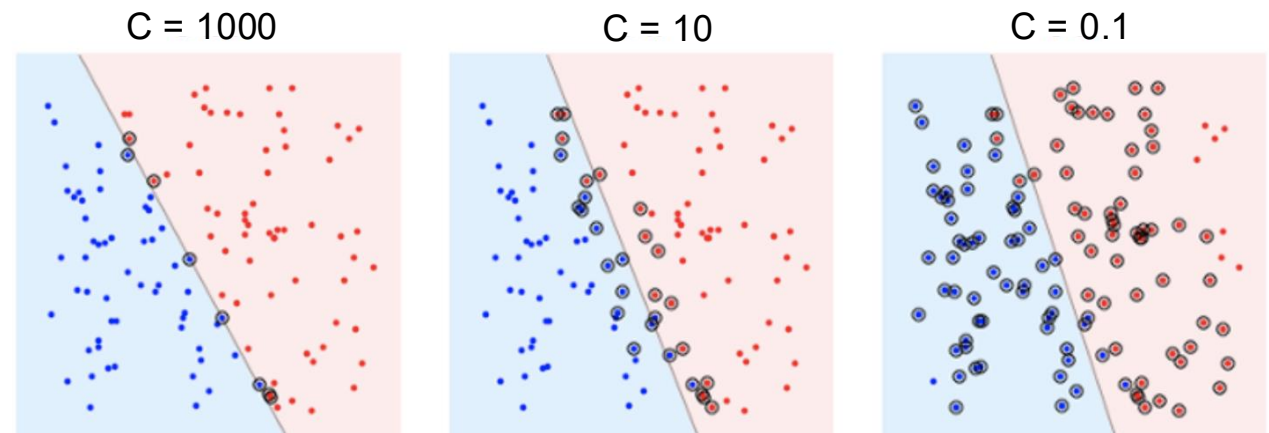
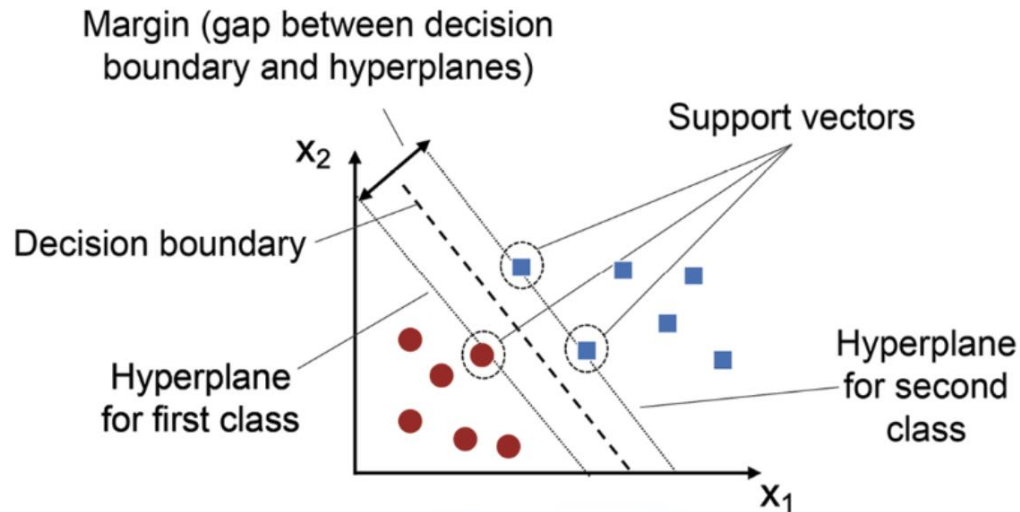
- Pairwise distances between conditions
- Discards raw patterns → geometry / representational structure
- Representational similarity matrices can be correlated with a model RDM (cross-modality correlations; consider noise-ceiling)

Decode individual fingers from activity patterns

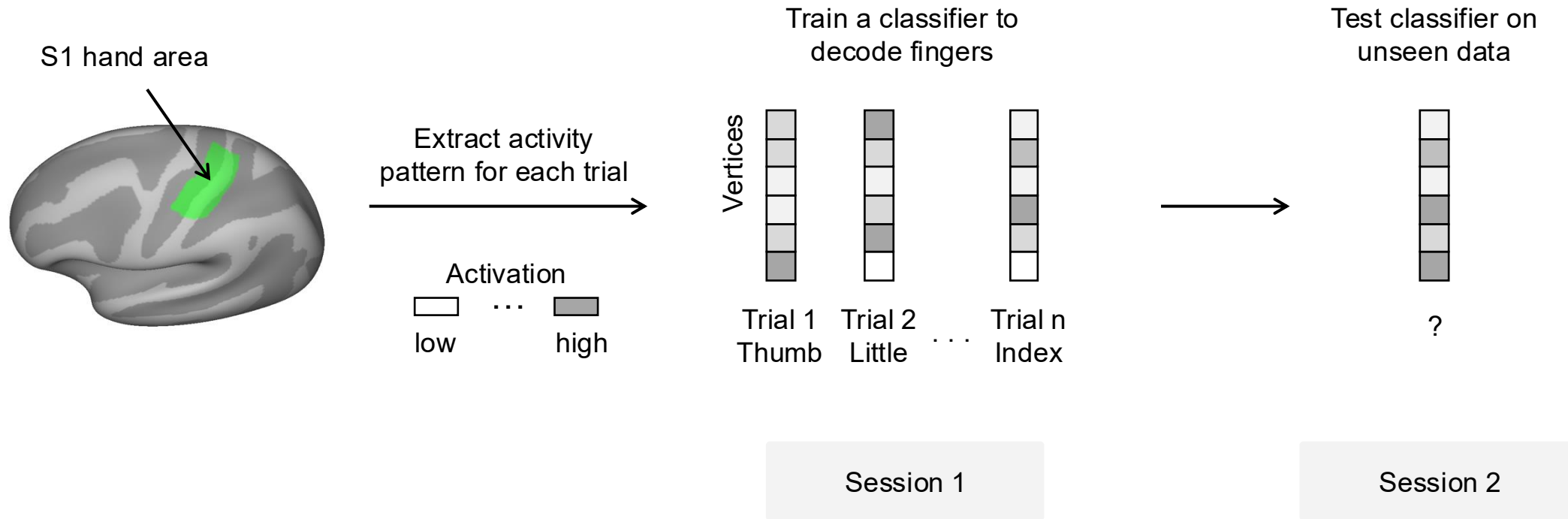


Choice of classifier

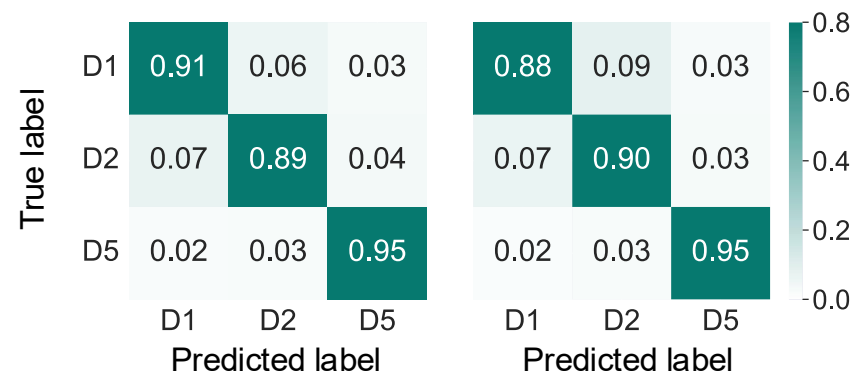
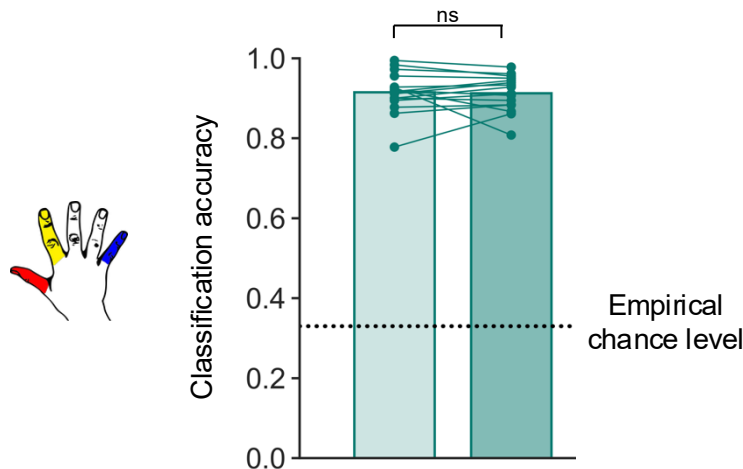
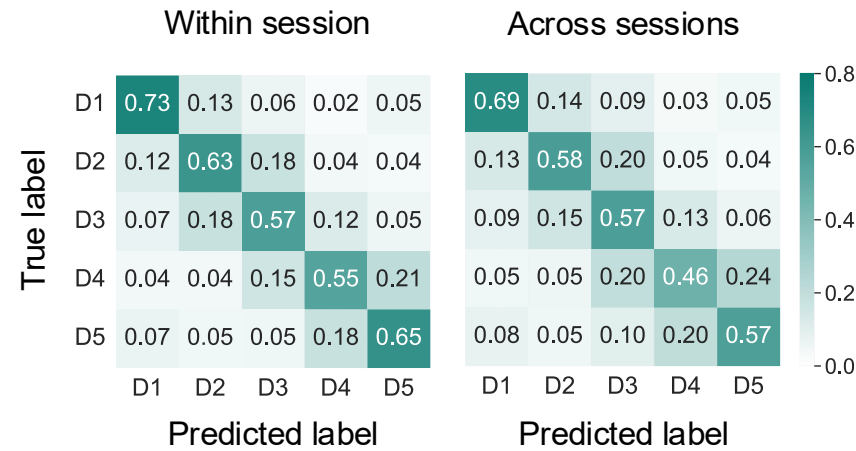
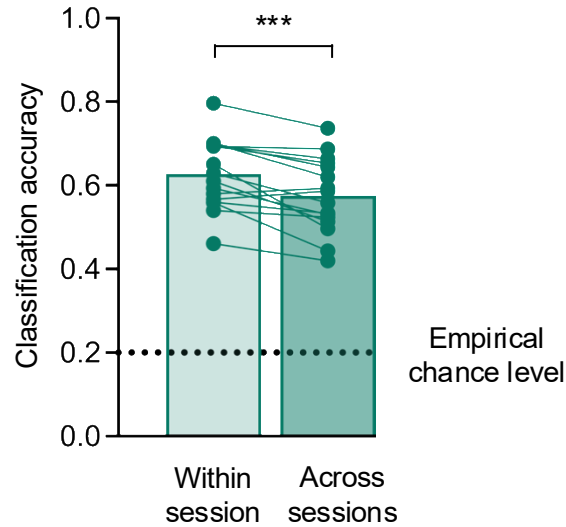
- In fMRI studies, we usually have many features (voxels or vertices) and few samples (trials or runs)
 - fMRI data is very noisy
- Linear SVMs can handle these issues quite well and are commonly used in fMRI studies
- Typically, we use it to test a hypothesis. The goal is not to tweak the classifier.



Investigate the stability of finger representations over time



Activity patterns associated with individual finger movements are stable over time in S1



Multivariate pattern analysis

Representational similarity analysis (RSA)

How is the information organised

- Pairwise distances between conditions
- Discards raw patterns → geometry / representational structure
- Representational similarity matrices can be correlated with a model RDM (cross-modality correlations; consider noise-ceiling)

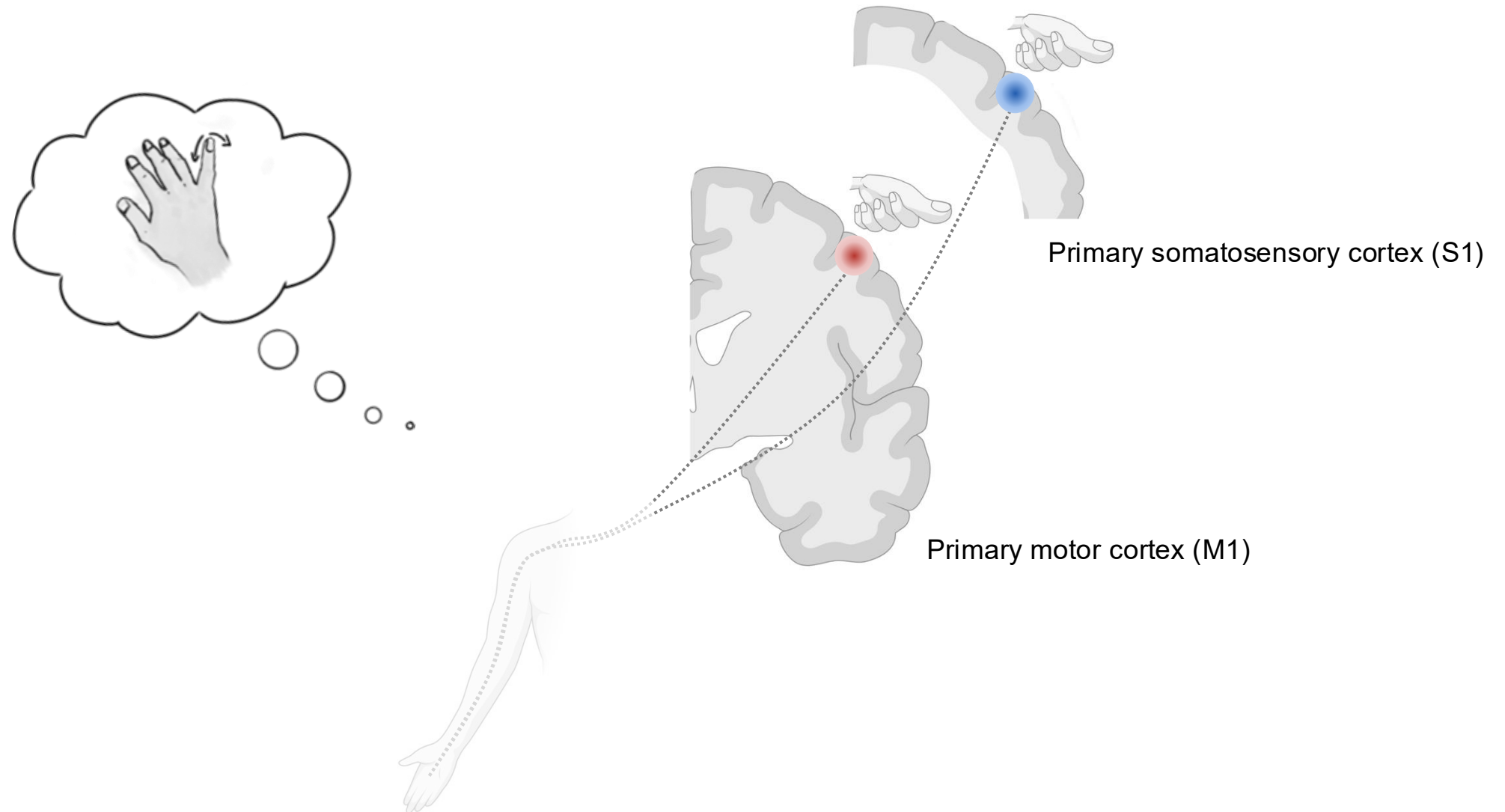
Decoding / Classification

Is the information present?

- Classification accuracy (and confusion matrices)
- Possible to analyse informative voxels
- Across-condition classification shows generalisation across tasks

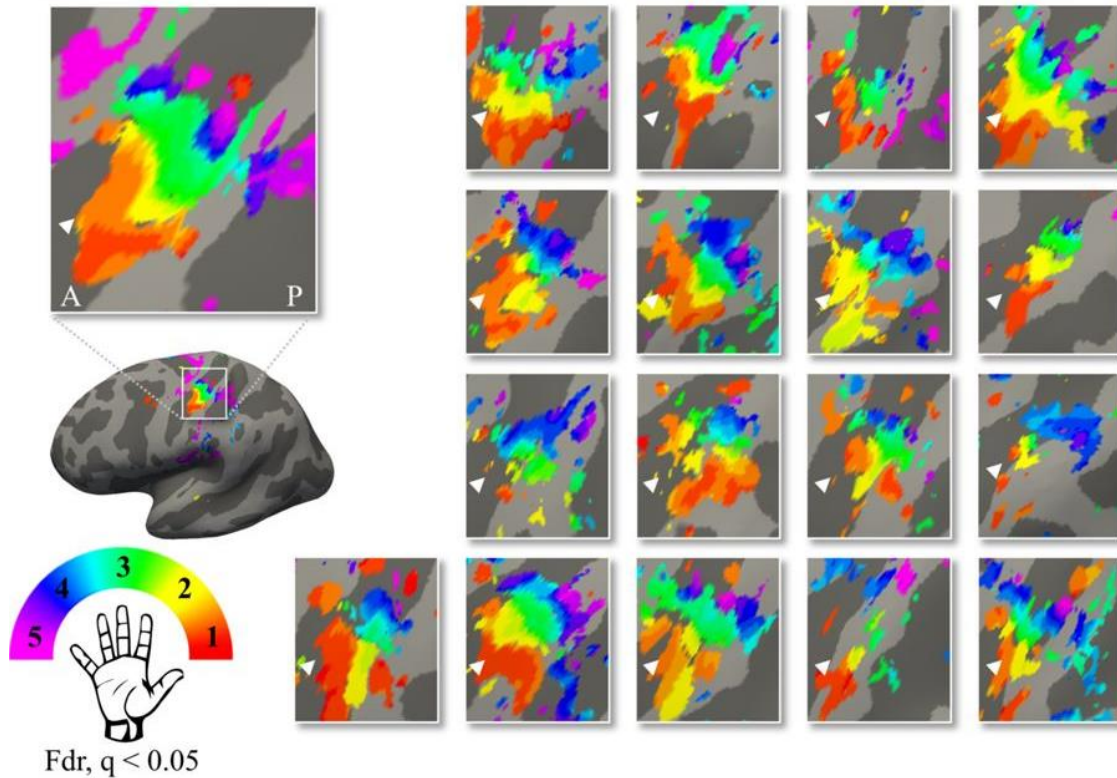
- Does not rely on overall activity level or thresholds
- In individual participant space (or average space for generalisation across participants)
- Region of interest vs search-light approaches

What if there is no motor output or somatosensory input?

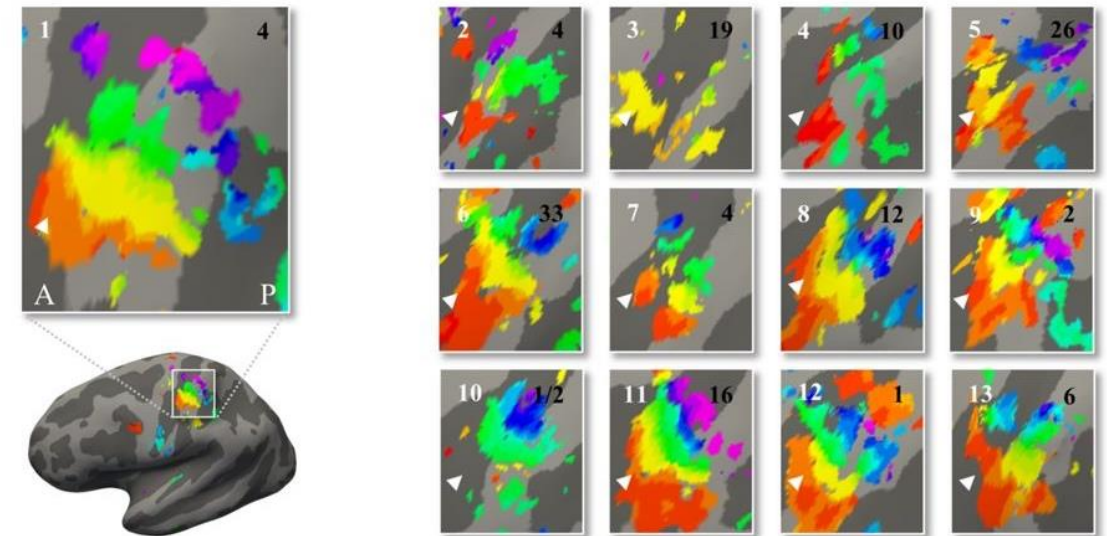


Attempted movements reveal preserved finger maps in tetraplegic participants after spinal cord injury

Able-bodied controls



Tetraplegic participants



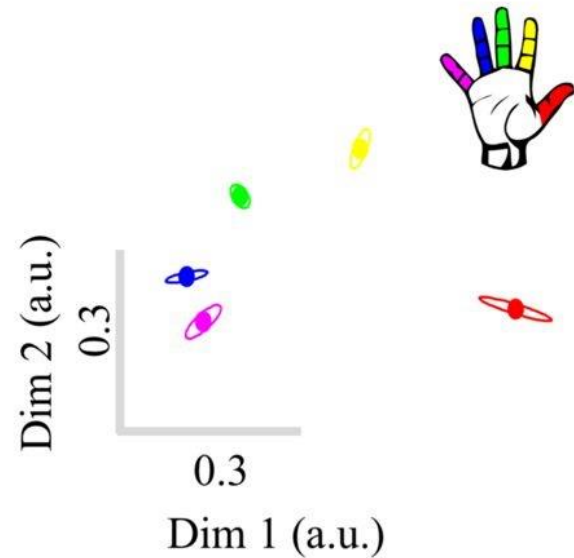
3T fMRI
2 mm³



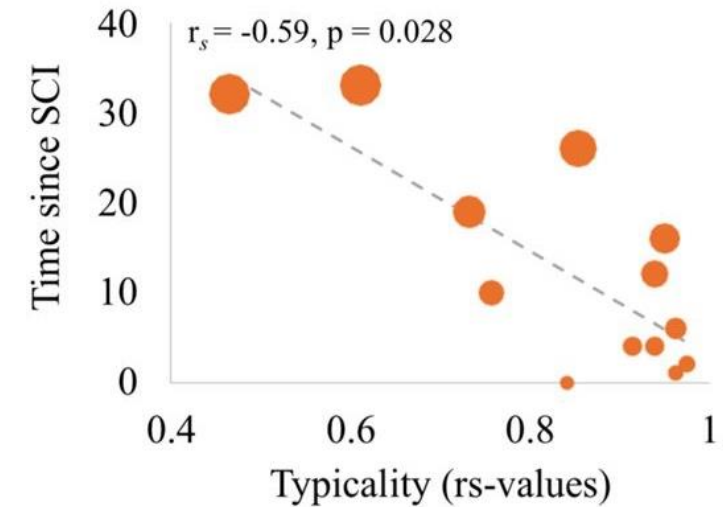
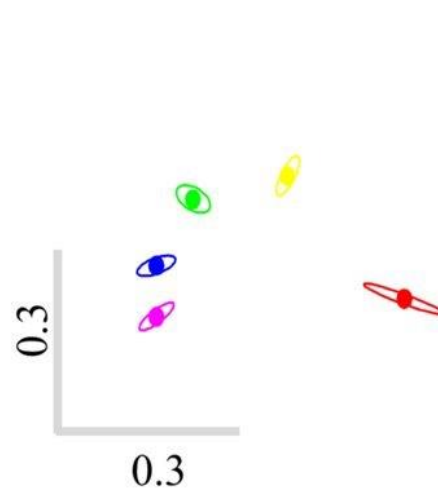
14 SCI
18 Controls

S1 finger somatotopy is preserved after tetraplegia but deteriorates over time

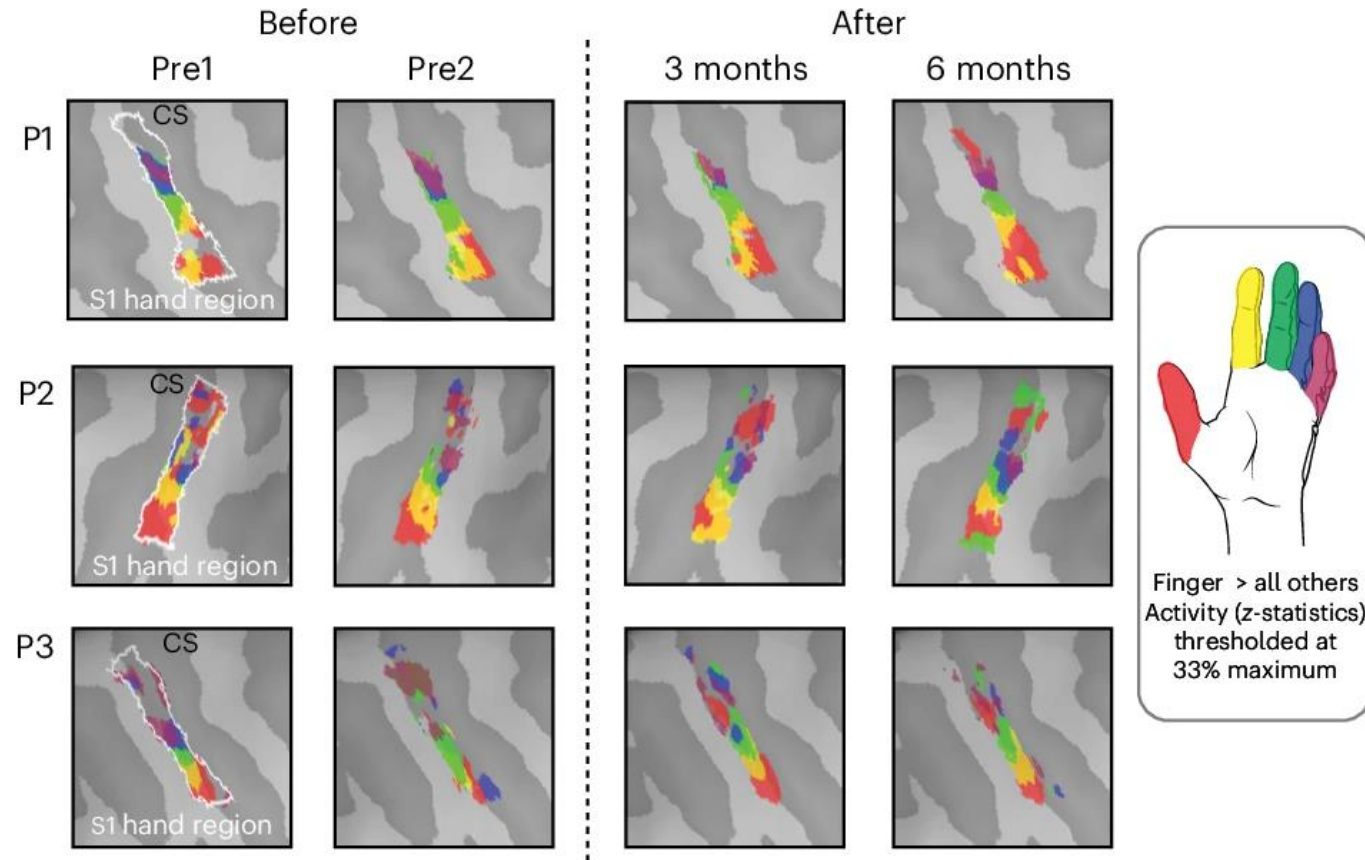
B) Hand representation controls



C) Hand representation patients



Phantom movements activate preserved finger maps after arm amputation

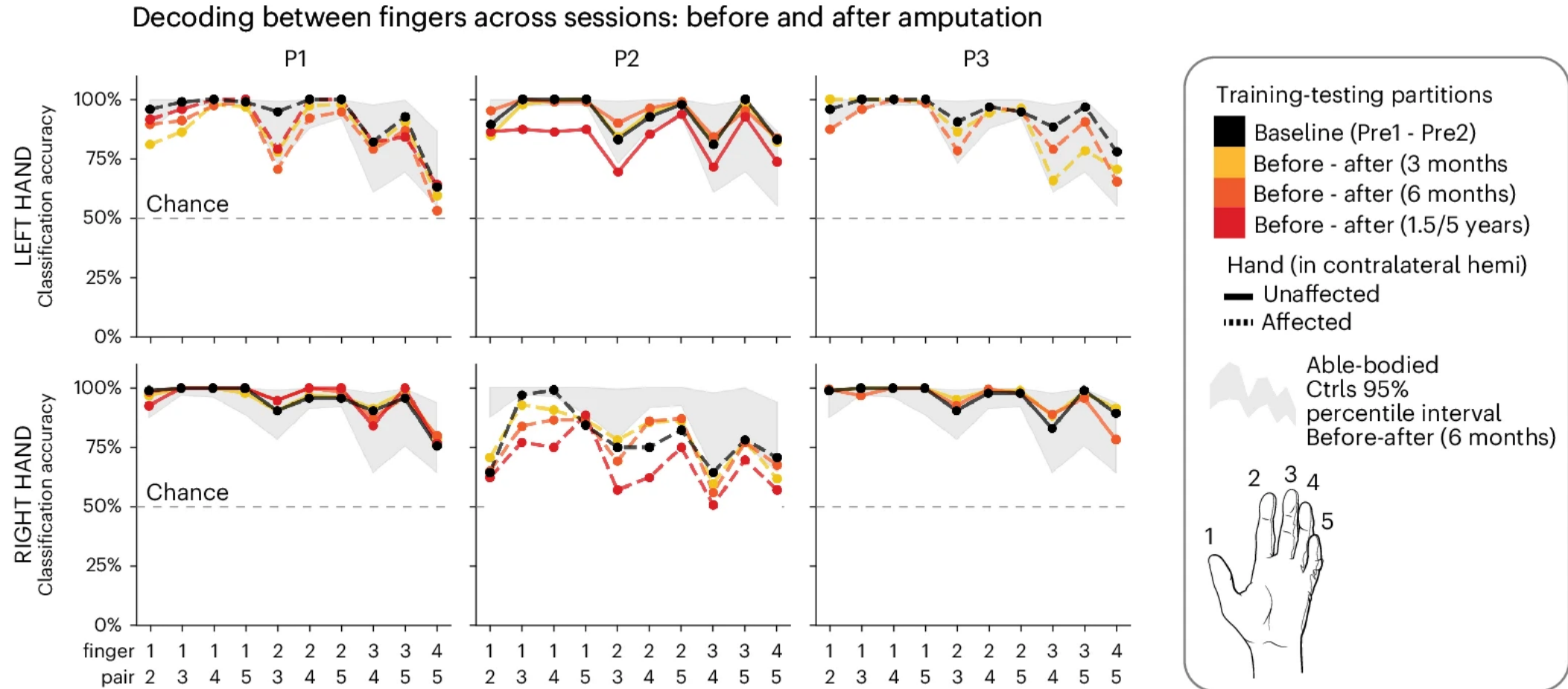


3T fMRI
2 mm³



3 Amputees
16 Controls

Finger selectivity decreases slightly after amputation



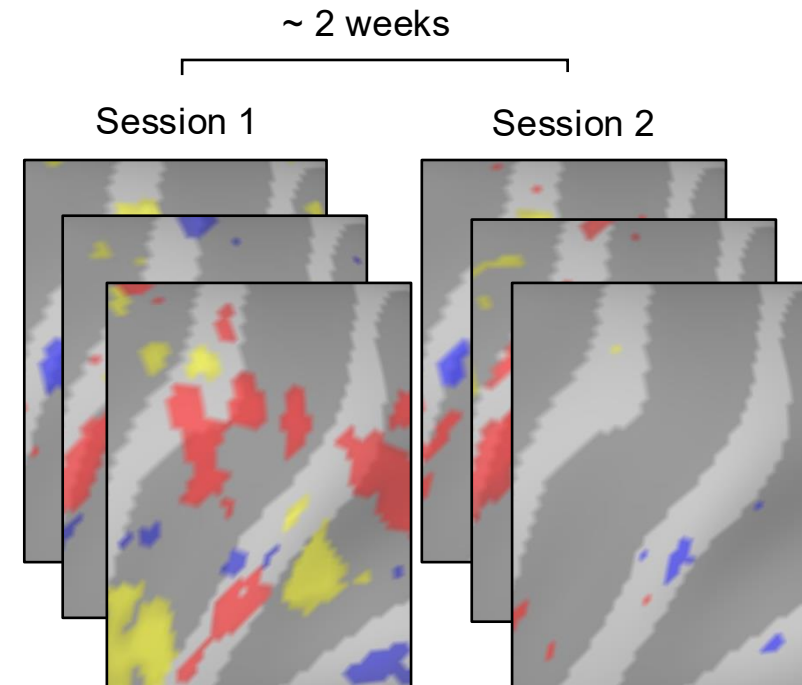
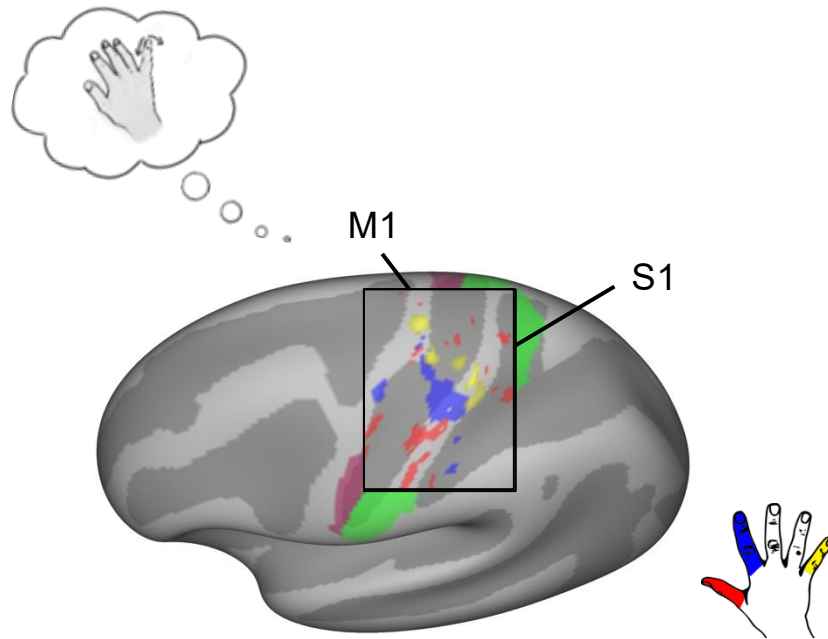
Motor imagery activates finger maps ... in most participants



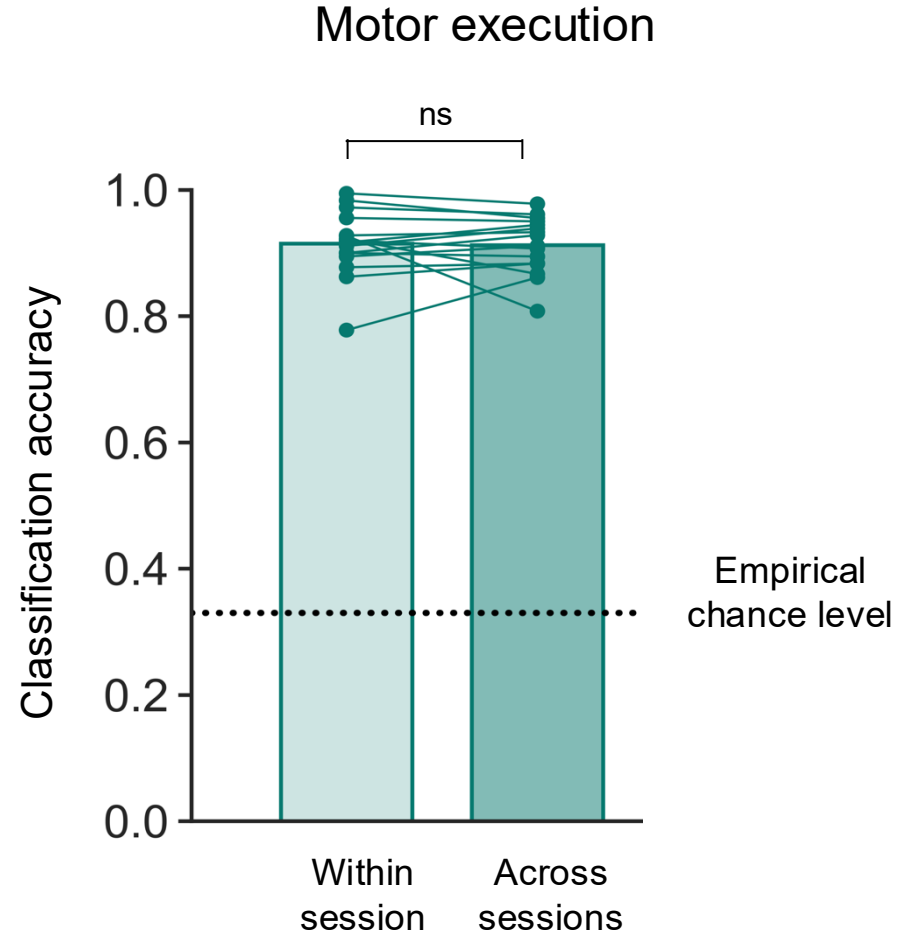
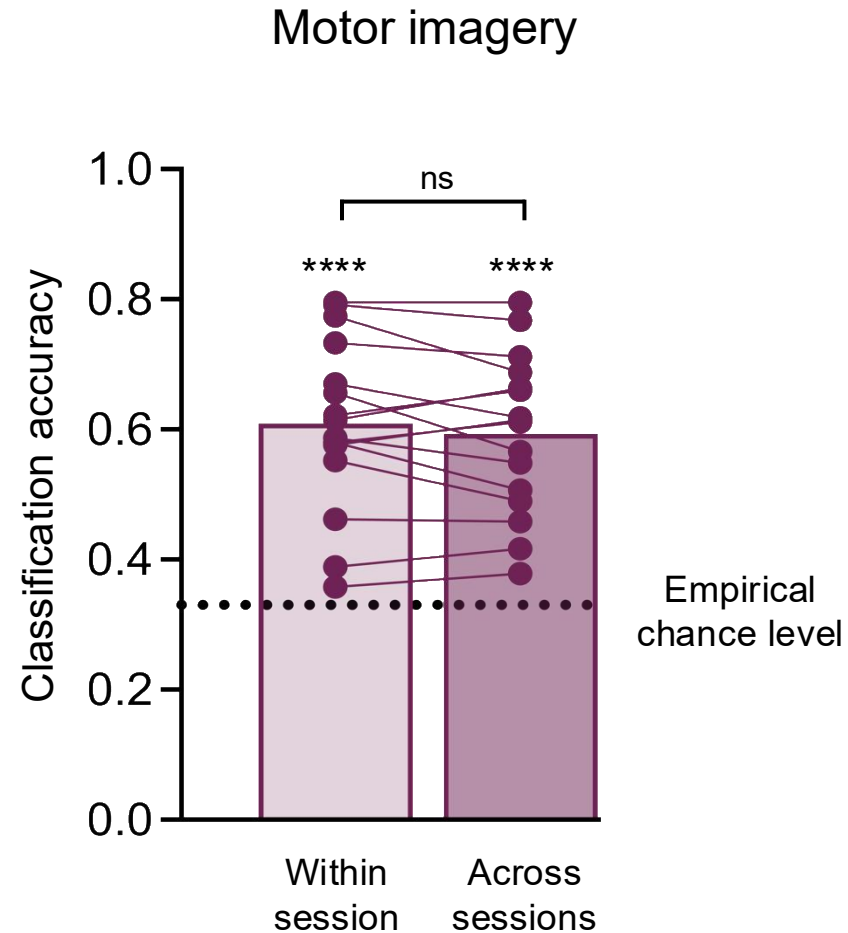
3T fMRI
2.2 mm³



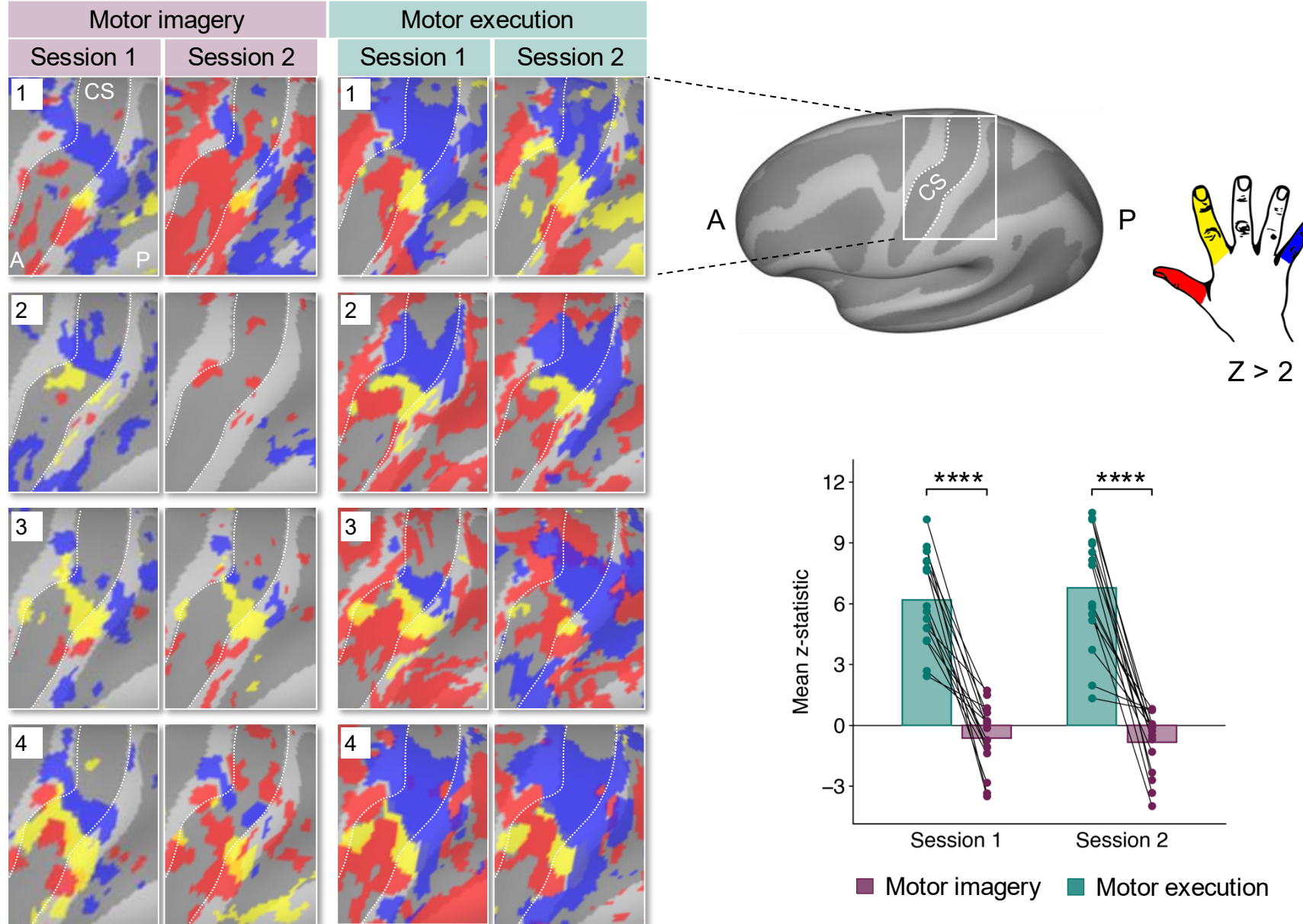
N = 16



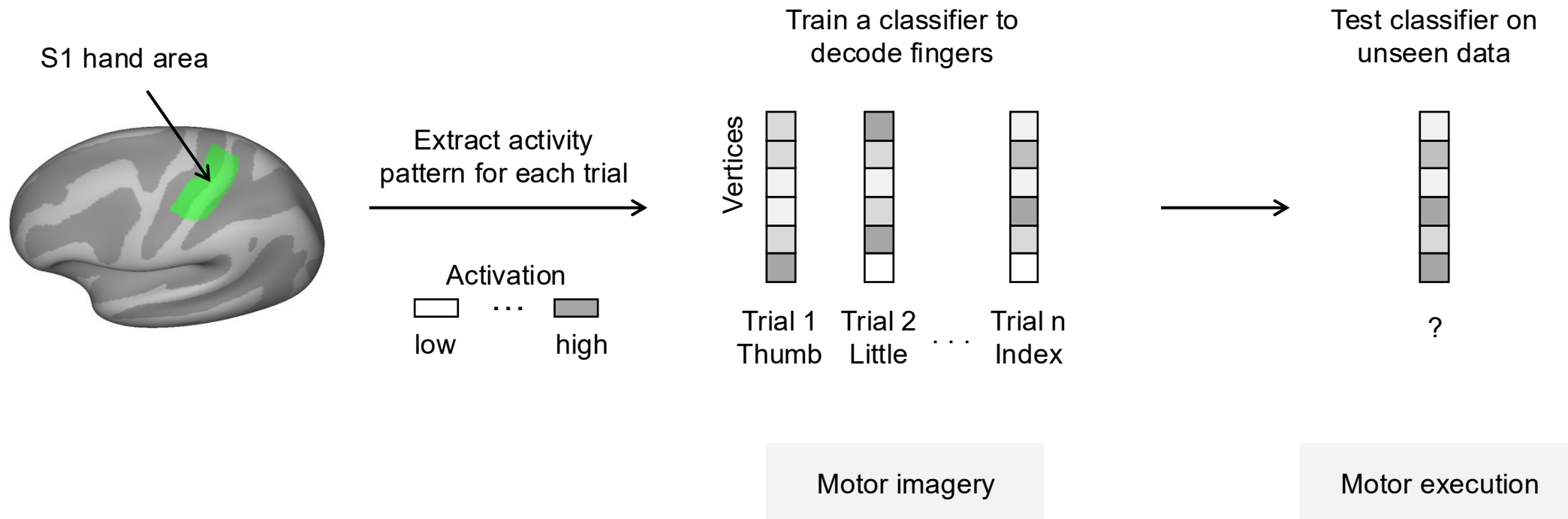
Imagined finger movements can be decoded from S1



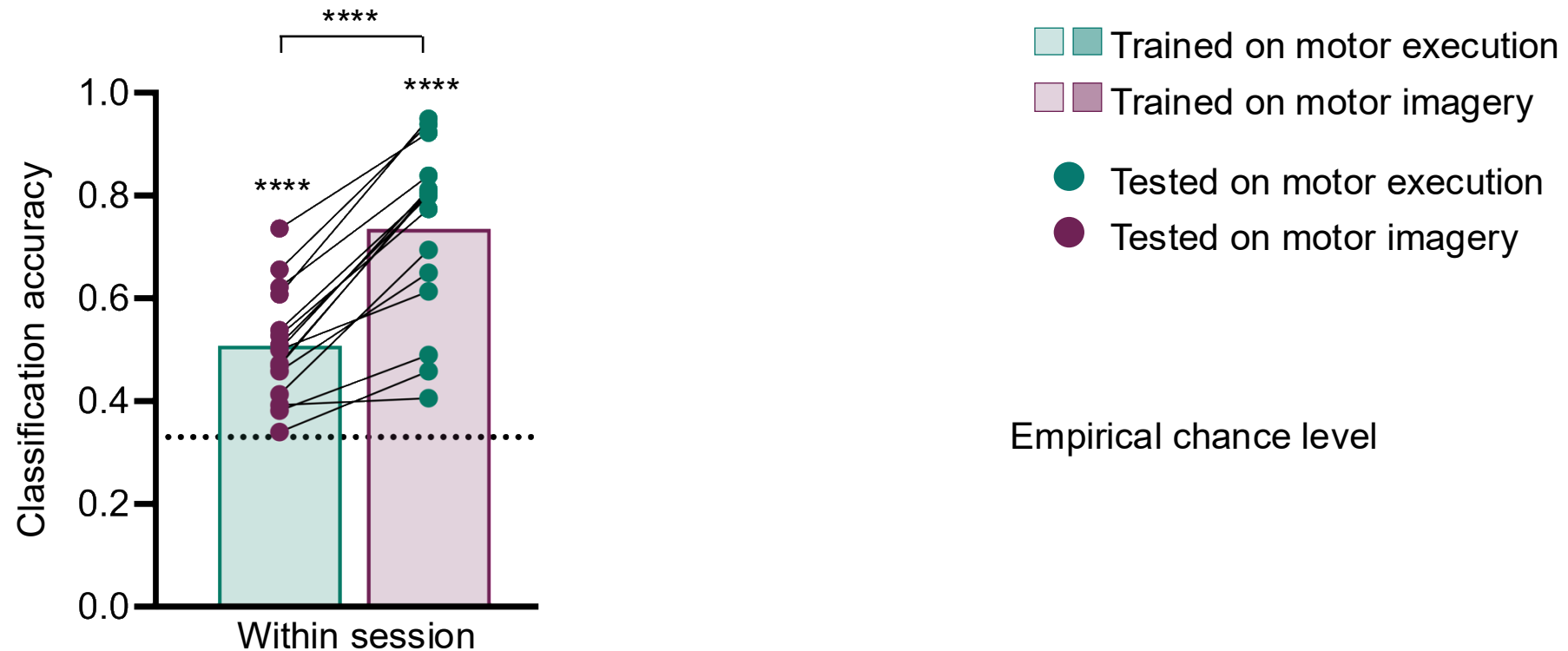
Does motor imagery activate the same finger representations?



Cross-task decoding analysis



Neural similarity of motor imagery and motor execution



Empirical chance level



Sensorimotor representations can be activated by **top-down tasks**

- even without somatosensory input and no motor output (i.e., no tactile / proprioceptive consequences):
 - Attempted movements
 - Phantom movements
 - Imagined movements
- to study sensorimotor representations in individuals with loss of sensorimotor function
 - Localise hand area for BCI implants
 - Leverage for BCI control
 - Use motor imagery-based neurofeedback as a training modality in neurorehabilitation that does not rely on overt movements